

# **Universal Control Network Planning**

**UN02-501**



**System Site Planning - 1**

***Universal Control  
Network  
Planning***

**UN02-501**

**Release 684**

**CE Compliant**

**March 2013**

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## About This Publication

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This publication provides the necessary information required to plan a Universal Control Network (UCN) or an Enhanced universal Control Network (EUCN).

This publication supports **TotalPlant** Solution (TPS) Network Release 684 and earlier software releases. TPS is the evolution of TDC 3000<sup>X</sup>.

This publication supports CE Compliant equipment. Any equipment designated as “CE Compliant” complies with the European Union EMC and its health and safety directives. All equipment entering the European countries after January 1, 1996 requires this type of compliance, denoted by the “CE Mark.”

|                  |                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ATTENTION</b> | <p>With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.</p> <p>Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms EHPM, ENIM, and EUCN, respectively.</p> |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Standard Symbols

## Scope

The standard symbols used in this publication are defined as follows.

### ATTENTION

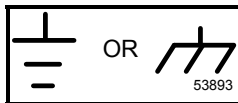
Notes inform the reader about information that is required, but not immediately evident.

### CAUTION

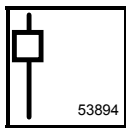
Cautions tell the user that damage may occur to equipment if proper care is not exercised.

### WARNING

Warnings tell the reader that potential personal harm or serious economic loss may happen if instructions are not followed.



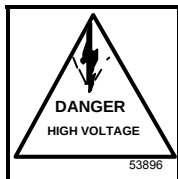
Ground connection to building safety ground.



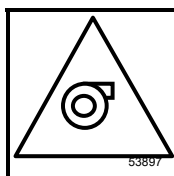
Ground stake for building safety ground.



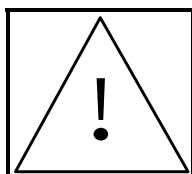
Electrical Shock Hazard—can be lethal.



Electrical Shock Hazard—can be lethal.



Rotating Fan—can cause personal injury.



Caution—refer to the appropriate installation document.

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# Acronyms

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|             |                                           |
|-------------|-------------------------------------------|
| APM .....   | Advanced Process Manager                  |
| APMM .....  | Advanced Process Manager Module           |
| AWG .....   | American Wire Gauge                       |
| CB .....    | Carrier Band                              |
| CF9 .....   | Control Firewall                          |
| EHPM .....  | Enhanced High Performance Process Manager |
| ENIM .....  | Enhanced Network Interface Module         |
| EPNI2 ..... | Enhanced Process Network Interface 2      |
| EUCN .....  | Enhanced Universal Control Network        |
| FTE .....   | Fault Tolerant Ethernet                   |
| HPM .....   | High-Performance Process Manager          |
| HPMM .....  | High-Performance Process Manager Module   |
| IEC .....   | International Electrotechnical Commission |
| ISO .....   | International Standards Organization      |
| LM .....    | Logic Manager                             |
| MAP .....   | Manufacturing Automation Protocol         |
| NEC .....   | National Electrical Code                  |
| NIM .....   | Network Interface Module                  |
| PM .....    | Process Manager                           |
| PMM .....   | Process Manager Module                    |
| SM .....    | Safety Manager                            |
| TPS .....   | <b>TotalPlant</b> Solution                |
| UCN .....   | Universal Control Network                 |
| UCNE .....  | Universal Control Network Extender        |
| UL .....    | Underwriters Laboratory                   |

## References

---

| <b>Publication<br/>Title</b>                                          | <b>Publication<br/>Number</b> | <b>TDC Binder<br/>Title</b>               | <b>Binder<br/>Number</b> |
|-----------------------------------------------------------------------|-------------------------------|-------------------------------------------|--------------------------|
| <i>TPS System Site Planning</i>                                       | SW02-550                      | System Site Planning -1                   | TPS 3020-1               |
| <i>Local Control Network Planning</i>                                 | SW02-501                      | System Site Planning -1                   | TPS 3020-1               |
| <i>LCN System Checkout</i>                                            | SW20-510                      | LCN Installation                          | TPS 3025                 |
| <i>LCN System Installation</i>                                        | SW20-500                      | LCN Installation                          | TPS 3025                 |
| <i>Logic Manager Installation</i>                                     | LM20-500                      | Implementation/Logic Manager              | TPS 3070                 |
| <i>Logic Manager Planning</i>                                         | LM02-501                      | System Site Planning -2                   | TPS 3020-2               |
| <i>Logic Manager Service</i>                                          | LM13-500                      | LM Service                                | TPS 3073                 |
| <i>Safety Manager Installation Guide</i>                              | SM20-500                      | Implementation/Safety Manager             | TPS 3074                 |
| <i>Safety Manager Service</i>                                         | SM13-500                      | Implementation/Safety Manager             | TPS 3074                 |
| <i>Process Manager/Advanced Process<br/>Manager Checkout</i>          | PM20-511                      | Implementation/PM/APM                     | TPS 3043                 |
| <i>Process Manager/Advanced Process<br/>Manager Installation</i>      | PM20-501                      | Implementation/PM/APM                     | TPS 3043                 |
| <i>Process Manager/Advanced Process<br/>Manager Planning</i>          | PM02-501                      | System Site Planning -1                   | TPS 3020-1               |
| <i>Process Manager/Advanced Process<br/>Manager Service</i>           | PM13-501                      | PM/APM/HPM Service -1                     | TPS 3061-1               |
| <i>High-Performance Process Manager<br/>Checkout</i>                  | HP20-510                      | Implementation/HPM -3                     | TPS 3066-3               |
| <i>High-Performance Process Manager<br/>Installation</i>              | HP20-500                      | Implementation/HPM 3                      | TPS 3066-3               |
| <i>High-Performance Process Manager<br/>Planning</i>                  | HP02-500                      | System Site Planning -2                   | TPS 3020-2               |
| <i>High-Performance Process Manager<br/>Service</i>                   | HP13-500                      | PM/APM/HPM Service -1                     | TPS 3061-1               |
| <i>Process Manager I/O Installation</i>                               | PM20-520                      | Implementation/PM/APM                     | TPS 3043                 |
| <i>Universal Control Network Guidelines</i>                           | UN12-510                      | Installation/Universal Control<br>Network | TPS 3041                 |
| <i>Universal Control Network Installation</i>                         | UN20-500                      | Installation/Universal Control<br>Network | TPS 3041                 |
| <i>Universal Control Network<br/>Specification and Technical Data</i> | UN03-500                      | System Summary - 2                        | TPS 3010-2               |



## Section 1 – Introduction

### 1.1 Overview

#### Section contents

These are the topics covered in this section:

|     | Topic            | See Page |
|-----|------------------|----------|
| 1.1 | Overview.....    | 1        |
| 1.2 | Definitions..... | 2        |

---

#### Purpose

This section describes the scope of the manual and lists support documents available from Honeywell.

---

#### Scope of this manual

This manual is used for planning the placement and installation of the Universal Control Network (UCN) at a site. The UCN provides a coaxial cable system that interconnects the Process Manager (PM), Advanced Process Manager (APM), High-Performance Process Manager (HPM), the Logic Manager (LM), Safety Manager (SM), and Network Interface Module (NIM) network devices.

---

#### What does “planning” cover?

Planning the Universal Control Network begins after the construction of the electronics building(s) in the process area and the design of the interior of the building(s). Discussion continues with other considerations such as power, grounding, wiring, and the equipment environment. The manual ends with design considerations of the UCN coaxial cable installation.

---

#### What is not covered?

The installation and checkout instructions, after power is applied to the Network Interface Module, Process Manager, Advanced Process Manager, High-Performance Process Manager, Logic Manager, or Safety Manager network devices, or the installation of the Local Control Network operator consoles are not covered in this manual. See the related documentation sections in the appropriate manuals for information concerning these topics.

---

#### Related documentation

See the References list at the front of this manual for a list of **TotalPlant** Solution (TPS) documents that are useful in the understanding and planning of the Universal Control Network.

---

## 1.2 Definitions

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>         | This section defines terms used in this document that may be unfamiliar to the normal TPS user.                                                                                                                                                                                                                                                                                                                             |
| <b>Network device</b>      | A network device is any electronic equipment that can be resident on the Universal Control Network. The equipment includes the Network Interface Module, the Process Manager, the Advanced Process Manager, the High-Performance Process Manager, the Logic Manager, and the Safety Manager. The Network Interface Module is always resident on the network because it provides the interface to the Local Control Network. |
| <b>UCN trunk cable</b>     | One of two coaxial cables, designated A or B, that provides the transmission media for the UCN. The trunk cable consists of segments connected together by UCN taps.                                                                                                                                                                                                                                                        |
| <b>UCN trunk segment</b>   | The portion of the UCN trunk cable separated by UCN taps.                                                                                                                                                                                                                                                                                                                                                                   |
| <b>UCN cable A</b>         | One of a redundant pair of UCN coaxial trunk cables that are diversely routed to reduce the possibility of simultaneous damage to the A and B cables.                                                                                                                                                                                                                                                                       |
| <b>UCN cable B</b>         | One of a redundant pair of UCN coaxial trunk cables that are diversely routed to reduce the possibility of simultaneous damage to the A and B cables.                                                                                                                                                                                                                                                                       |
| <b>Drop cable</b>          | The coaxial cable that connects the network device to the UCN trunk cable (UCN Cable A or UCN Cable B) by means of the trunk cable tap.                                                                                                                                                                                                                                                                                     |
| <b>Trunk cable tap</b>     | The component that adapts the Network Device to the UCN trunk cable by means of a drop cable while maintaining the impedance of the trunk cable. Three types of UCN taps provide connections for two, four, or eight drop cables.                                                                                                                                                                                           |
| <b>Tap port</b>            | An F-type connector on the trunk cable tap that provides a connection for either a trunk cable or a drop cable.                                                                                                                                                                                                                                                                                                             |
| <b>UCN Extender (UCNE)</b> | A fiber optic link between coaxial trunk cable segments.                                                                                                                                                                                                                                                                                                                                                                    |
| <b>UCNE</b>                | The device that provides the fiber optic link between coaxial trunk cable segments. A pair of UCNEs are required at each end the fiber optic link.                                                                                                                                                                                                                                                                          |

## Section 2 – Universal Control Network

### 2.1 Overview

---

#### Section contents

These are the topics covered in this section:

|      | Topic                                | See Page |
|------|--------------------------------------|----------|
| 2.1  | Overview.....                        | 3        |
| 2.2  | UCN Hardware Components .....        | 4        |
| 2.3  | Connectors .....                     | 5        |
| 2.4  | UCN Taps.....                        | 6        |
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| 2.10 | Connector Torque Wrench .....        | 16       |

---

#### Introduction

This section describes the communications network used by the Network Interface Module (NIM), Process Manager (PM), Advanced Process Manager (APM), High-Performance Process Manager (HPM), Logic Manager (LM), and Safety Manager (SM) and its installation.

---

#### What is the Universal Control Network?

Communication from the control room to the PM, APM, HPM, LM, and SM is accomplished through a coaxial cable system. This system and its protocol is designated the Universal Control Network (UCN). The UCN uses standard ISO 802.4 hardware. Figure 2-1 illustrates a typical UCN installation.

---

#### Planning considerations

To plan a UCN for a site, consider the following:

- Choose UCN trunk cable routes that bypass sources of electrical interference and are easy to implement.
- Wherever possible, use separate routes for the two UCN trunk cables.
- Choose trunk cables to suit the route distance, number of connected network devices, and other special requirements.
- Plan the placement of UCN taps, splices, and other components needed to interconnect the UCN devices.

---

#### No trunk cables

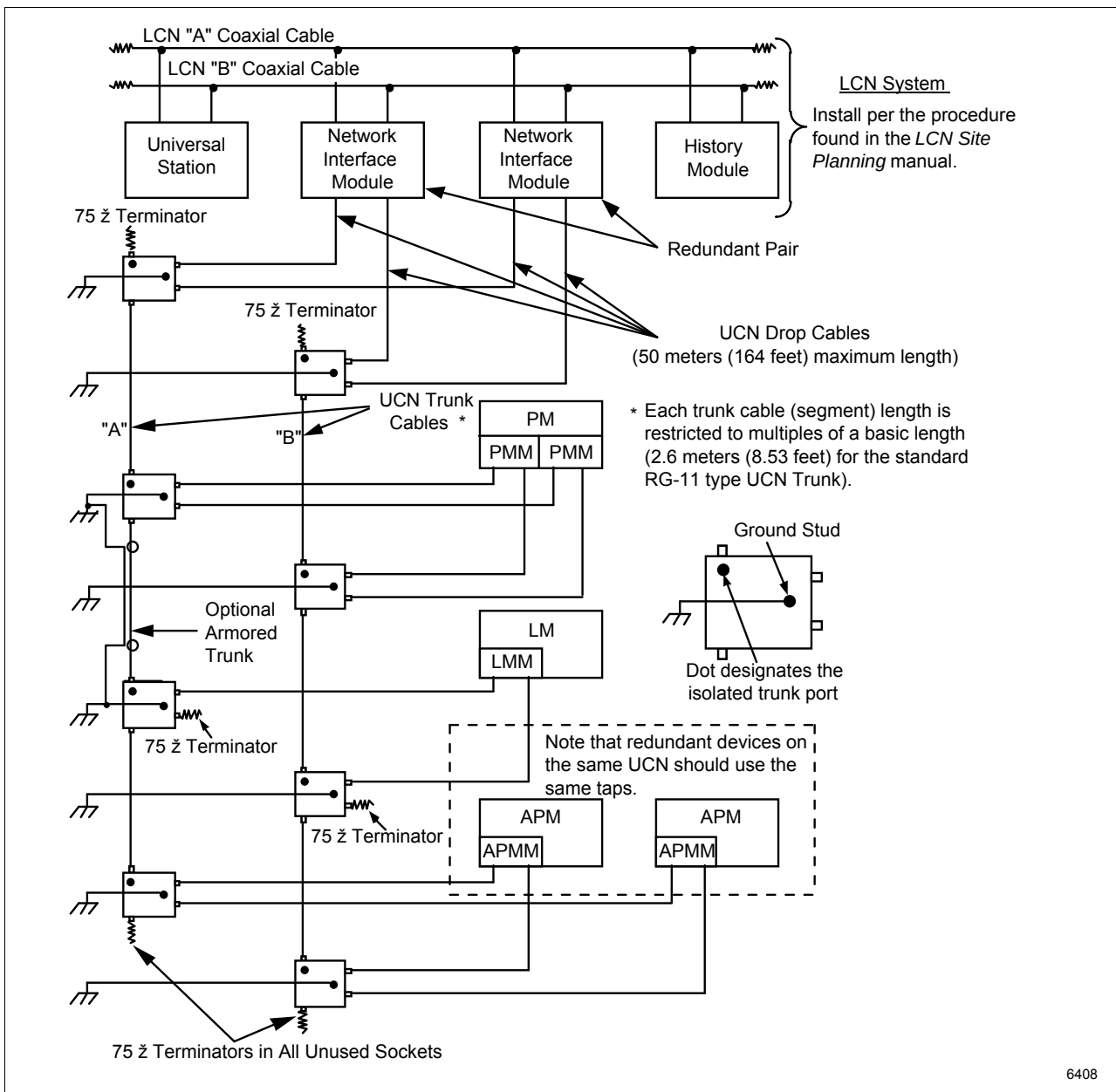
It is acceptable to construct a Universal Control Network that consists of one Cable A drop tap and one Cable B drop tap, with drop cables connected to UCN node devices and no trunk cables. The drop taps' trunk ports must be terminated.

## 2.2 UCN Hardware Components

### Introduction

The various hardware components that compose the UCN Network are defined in the text that follows. Where possible, the components are also illustrated in diagrams. Figure 2-1, a schematic of a typical UCN installation, can be used to further identify the components.

Figure 2-1 Typical Universal Control Network Installation



6408

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|                          |                                                                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Where is the UCN?</b> | Two UCN trunk cables (shown vertically in Figure 2-1 along the left side) pass from one end of the network to the other. The trunks are terminated at each end with a 75Ω terminator. Taps are added where “drops” connect TPS equipment to the trunks. |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## 2.3 Connectors

---

|                              |                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------|
| <b>The type of connector</b> | All UCN connectors are industrial-grade F-type with gold-plated center pins or sockets. |
|------------------------------|-----------------------------------------------------------------------------------------|

---

### **WARNING**

WARNING—The F-type connector on the UCN drop cables used in an FSC-SM node is replaced by a new F-connector (a component of the cable assembly bearing part number: 4010242) with a revised torque setting.

UCN drop cables with the new F-type connectors must be tightened to a torque of 12 -15 inch/pounds.

The new type connectors must not be tightened to original torque setting as this may damage the connector.

Newly ordered cables have a warning that specifies the new torque setting.

---

|                             |                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Corrosion resistance</b> | The Honeywell F-type connector is unique because it uses a corrosion-resistant center pin. This contrasts with the commercial practice of using the coax’s center conductor as a connector pin. |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

|                               |                                                                                                                                                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Connector part numbers</b> | Connectors are supplied in kits. Part numbers of connector kits designed to fit a specific cable type are referenced in the service manual for the particular UCN device you are installing and in the appendix of this manual. |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

|                                         |                                                                                                               |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Connector installation procedure</b> | Installation of the connectors is detailed in the installation manuals for the PM, APM, HPM, LM, SM, and LCN. |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|

---

## 2.4 UCN Taps

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What do UCN taps do?</b>        | The UCN taps contain UCN signal transformers and impedance matching networks. These components allow the drop cables to interface to the trunk cable and also isolate the operating trunk from faults on the drop cables, such as shorts.                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Where are UCN taps located?</b> | UCN taps install in redundant pairs, one for the A cable and one for the B cable. They can be located up to 50 cable-meters (164 cable-feet) from the PM, APM, HPM, LM, SM, or NIM. Cabinets from the factory have the taps mounted in the cabinets but taps can be mounted elsewhere, such as above the ceiling tile or under the raised floor in the electronics room as long as standard grounding and other requirements are met. It is recommended that trunk cables and taps be located in a secure environment to eliminate trunk or tap faults due to physical damage, faulty connections, loose terminators, etc. |
| <b>UCN taps</b>                    | There are three types of UCN taps used in the Universal Control Network: 2-drop, 4-drop, and 8-drop. See Figure 2-2 for illustrations of these taps and the Section 7 of this manual for part numbers.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>2-Drop trunk cable tap</b>      | The 2-drop trunk cable tap provides for connection of two nonredundant devices or one redundant device to the trunk cable using two drop cables. The network shown in Figure 2-1 uses all 2-drop UCN taps to connect two redundant NIMs to a redundant PM, two redundant APMs, and a nonredundant LM.                                                                                                                                                                                                                                                                                                                      |
| <b>4-and 8-Drop UCN taps</b>       | More equipment drops in the same area are accomplished by using these larger cable taps. The 4-drop trunk cable tap accommodates four drop cables and the 8-drop trunk cable tap accommodates eight drop cables.                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>How are UCN taps supplied?</b>  | UCN taps are shipped as pairs in kits that contain two 2-drop UCN taps with four 75 $\Omega$ terminators, two 4-drop UCN taps with eight 75 $\Omega$ terminators, or two 8-drop UCN taps with sixteen 75 $\Omega$ terminators. Each kit also includes mounting brackets that can be used to install the taps in a cabinet or elsewhere.                                                                                                                                                                                                                                                                                    |

## 2.4 UCN Taps, Continued

---

**How are UCN taps grounded?**

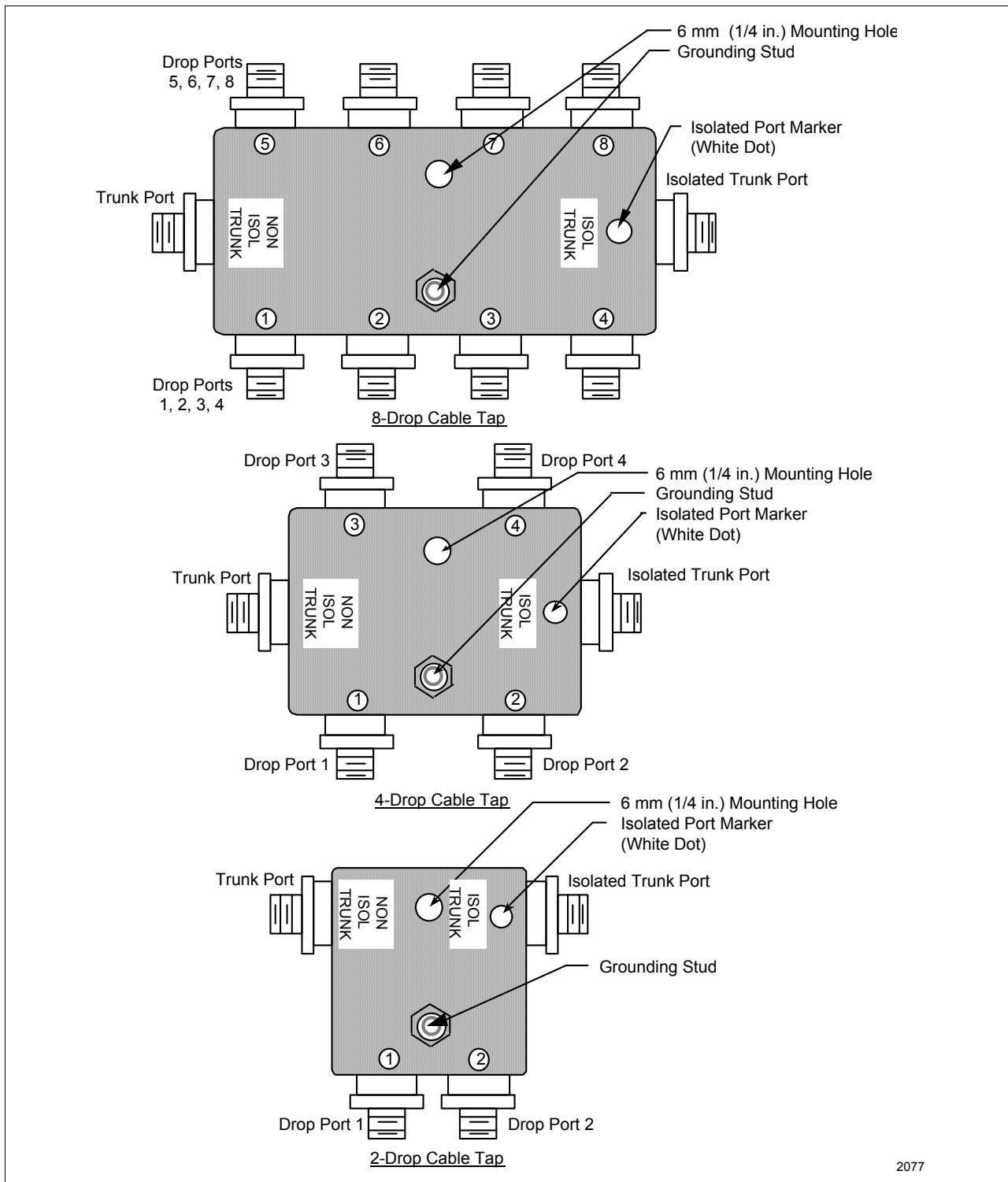
Each trunk cable tap must be connected to a local ac safety ground with a 5.0 mm<sup>2</sup> (10 AWG) green or bare conductor (or a flat braid equivalent) fastened to the tap's ground stud as illustrated in Figures 2-1 and 2-2. Do not daisy-chain these grounds or use the ground stud to mount the tap. Use the mounting hole in the body of the tap.

Each tap has one trunk connection that grounds the trunk cable shield and one connection that isolates the trunk cable shield. A trunk cable between two successive taps must be connected to an isolated trunk port on one end and a grounded trunk port on the other end. Refer to Figure 2-1.

---

## 2.4 UCN Taps, Continued

Figure 2-2 Universal Control Network UCN Taps



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## 2.5 Terminators

---

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Terminator installed</b>                   | <p>All unused UCN cable or trunk cable tap connectors must have a 75-ohm UCN terminator connected to it that is wrench-tightened to 25 inch-pounds.</p> <p>This includes the terminators at both ends of the trunk, as well as the terminators connected to each unused trunk cable tap connector.</p>                                                                                                                                                                                                                                                                                                    |
| <b>Reasons for termination</b>                | <p>The terminator maintains proper system impedance and provides connector protection against corrosion.</p> <p>The UCN terminator is specifically designed for carrier band use. Commercial or broadband terminators are not acceptable for the Universal Control Network.</p>                                                                                                                                                                                                                                                                                                                           |
| <b>Reasons for termination and tightening</b> | <p>Proper termination and connector tightening are needed for the following reasons.</p> <ul style="list-style-type: none"><li>• Loose terminators can produce UCN noise which will degrade UCN performance.</li><li>• End of trunk terminators must be tightened with a torque wrench to provide correct tightness. See subsection 2.10 for details.</li><li>• The metal chain/ring attached to terminators or the terminator body itself, must not be allowed to touch any nearby metal surface other than the tap port connector itself. UCN noise could result and degrade UCN performance.</li></ul> |

---

## 2.6 Cable Splices

|                            |                                                                                                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reason for splices</b>  | The primary use of a splice is to connect two or more trunk cable segments if a longer trunk cable segment is needed. The splice must be mechanically sound and must protect the cable components from the elements. |
| <b>F-type splice</b>       | The splice is a female F-type to female F-type union that allows connection of cables with a male F-type connector at each end.                                                                                      |
| <b>Splice kit</b>          | The Honeywell UCN splice kit (see the appendix for part number) contains ten splices, a roll of gel-lined sealing tape, and heat shrink tubing for mechanical protection.                                            |
| <b>Splice installation</b> | Wrench-tightening the connectors to 25 inch-pounds of torque, then seal the connections at the splice, using heat shrink tubing and gel-lined sealing tape.                                                          |

## 2.7 Coaxial Cables

### Description

The Universal Control Network uses coaxial cables throughout the system.

The smaller diameter drop cables are for tapping into the trunk through a trunk cable tap at each desired location, while the larger diameter trunk cables carry the signal along the network. The cables are discussed in the following subsections.

## 2.8 Standard RG6-Type Drop Cable

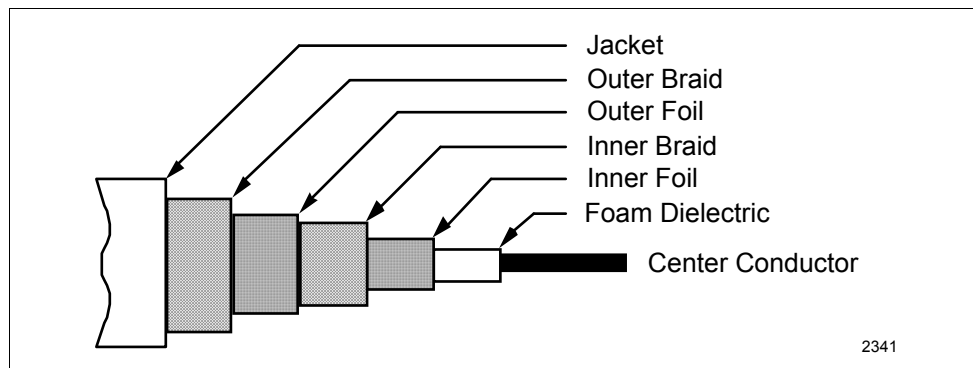
### CAUTION

UCN coaxial cables and associated components have been designed and tested by Honeywell to meet strict standards of performance. Other coaxial cables and components have not been tested and qualified for use in the UCN; therefore, no substitutions are approved by Honeywell. Various lengths of coaxial cable are available as standard products and must be ordered from Honeywell or a Honeywell approved sources.

### Drop cable structure

Standard indoor UCN drop cable is a quad-shielded RG6-type coaxial cable as illustrated in Figure 2-3.

Figure 2-3 RG6-Type Coax Drop Cable Structure



### Drop cable specifications

Drop cable is approximately 8 mm (5/16 inches) in diameter, has a minimum bend radius of 8 cm (3 inches), and a maximum pull-strength, without permanent deformation, of 45 kg (100 pounds).

Its maximum rated working distance is 50 cable-meters (164 cable-feet) from the trunk cable tap to the UCN device (PM, APM, HPM, LM, SM, or NIM). Longer runs are not supported by the Manufacturing Automation Protocol (MAP) Carrier Band (CB) standards and should not be used.

Sources of approved cable are discussed in the following text.

|                                                 |                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Drop cable listings and ratings</b>          | Standard UCN drop cable has an Underwriters Laboratory (UL) listing as a Class 2 flame retardant cable. See your Honeywell sales engineer for the latest information on plenum-rated or other drop cable.                                                                                                            |
| <b>Importance of routing</b>                    | The routing of coaxial cables is important allowing for the probability that if one cable is susceptible to electrical noise or physical damage the other cable will likely remain intact and operational. See the power line separation rules in subsection 3.3. Also, see subsection 6.2 on importance of routing. |
| <b>Factory manufactured sets</b>                | Standard UCN drop cables are available from Honeywell in manufactured sets of various lengths of up to 50 meters. See the parts list in the service manual for the particular UCN device you are installing.                                                                                                         |
| <b>Factory manufactured markings</b>            | A set of drop cables consists of an A cable and a B cable. Each end of a factory manufactured drop cable has a marker identifying the cable as cable A or B. The A cable markers are orange and the B cable markers are brown.                                                                                       |
| <b>Field fabricated sets</b>                    | If required, drop cables can be fabricated on site using bulk cable on a reel, connector kits, and crimping tools available from Honeywell. Only Honeywell recommended cable must be used.                                                                                                                           |
| <b>Drop cable in bulk</b>                       | Standard UCN drop cable can be ordered in various lengths on a reel from Honeywell under part number 51190898-xxx, where -xxx designates the cable length ordered.                                                                                                                                                   |
| <b>Field installation of drop cable markers</b> | Identification markers must be installed on field fabricated cables. Failure to do so may result in poor performance and lost time in troubleshooting.                                                                                                                                                               |
| <b>Testing of fabricated cables</b>             | Any fabricated coaxial cable should be tested as described in the <i>Universal Control Network Installation</i> manual before it is installed in the UCN. If the drop cable is pulled through conduit or cable tray, it should be tested before and after pulling.                                                   |
| <b>Connectors and tools</b>                     | <p>Connectors for RG6-type drop cable can be ordered from Honeywell as a kit with 24 connectors. The kit also includes a supply of brown and orange cable markers.</p> <p>A drop cable connector crimping tool is available in a kit from Honeywell. See the Section 7 in this manual for the kit part number.</p>   |

## 2.9 Standard RG11-Type Trunk Cable

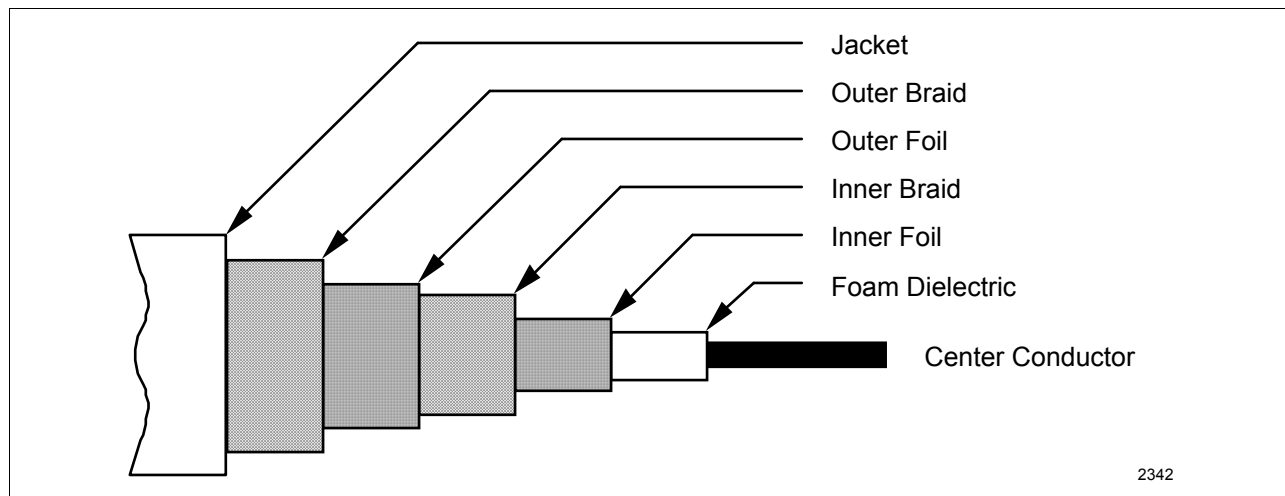
### CAUTION

CAUTION—UCN coaxial cables and associated components have been designed and tested by Honeywell to meet strict standards of performance. Other coaxial cables and components have not been tested and qualified for use in the Universal Control Network; therefore, no substitutions are approved by Honeywell. Various lengths of coaxial cable are available as standard products and must be ordered from Honeywell or Honeywell approved sources.

**Trunk cable structure** Trunk cables interconnect the UCN taps as illustrated in Figure 2-1.

Standard UCN indoor trunk cable is a special quad-shield RG11-type coaxial cable, designed for Manufacturing Automation Protocol (MAP) Carrier Band (CB) applications such as the UCN. Figure 2-4 is an illustration of the cable.

Figure 2-4 RG11-Type Coax Drop Cable Structure



### Trunk cable specifications

Standard UCN indoor trunk cable is approximately 11 mm (7/16 inches) in diameter, has a minimum bend radius of 11 cm (4 1/2 inches), and a maximum pull-strength, without permanent deformation, of 68 kg (150 pounds). Trunk cable length limitations are discussed in subsection 3.2.

### Trunk cable listings and ratings

Standard indoor UCN trunk cable has a UL listing as a Class 2 flame retardant cable.

Other trunk cables can be plenum rated, armored, rigid, semi-rigid, or for direct burial. See your Honeywell sales engineer for the latest information. The use of armored trunk cable or the placement of trunk cable in conduit can be required in some installations to ensure proper performance of the UCN when unacceptable levels of electrical noise are present.

|                                                  |                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Importance of routing</b>                     | The routing of a coaxial cable is important. Power line separation rules are discussed in subsection 3.3.                                                                                                                                                                                                                                                          |
| <b>WARNING</b>                                   | <p>WARNING—Standard indoor UCN trunk cable with a PVC jacket is not recommended for outdoor use unless it is installed in waterproof conduit. The effects of sunlight and weather cause PVC-jacketed cable to deteriorate prematurely.</p> <p>An outdoor polyethylene-jacketed UCN trunk cable is available in bulk form. See “Outdoor cable available” below.</p> |
| <b>Manufactured sets of trunk cables</b>         | RG11-type trunk cables are available from Honeywell in manufactured sets of various lengths. See the parts list in the appendix of this manual for the particular UCN device you are installing.                                                                                                                                                                   |
| <b>Trunk cable markers</b>                       | Manufactured trunk cable sets consists of an A cable and a B cable. Each end of the trunk cable has a marker identifying the cable as cable A or B. The A cable markers are orange and the B cable markers are brown.                                                                                                                                              |
| <b>Field fabrication of trunk cables</b>         | Trunk cables can be fabricated on site if required. Only Honeywell recommended cable must be used.                                                                                                                                                                                                                                                                 |
| <b>Trunk cable in bulk</b>                       | Standard indoor UCN trunk cable without connectors attached to the ends can be ordered in various lengths on reels from Honeywell under part number 51190899-xxx, where -xxx is the cable length ordered.                                                                                                                                                          |
| <b>Field installation of trunk cable markers</b> | Cable markers that identify the cable as “A” or “B” must be installed on field fabricated cables. Failure to do so may result in poor performance and lost time in troubleshooting.                                                                                                                                                                                |
| <b>Outdoor cable available</b>                   | An outdoor-rated polyethylene-jacketed UCN trunk cable, Honeywell part number 51191607-xxx, is available as bulk cable on a reel for cable assemblies to be fabricated in the field and installed outdoors.                                                                                                                                                        |

---

**Performance considerations**

To maximize trunk cable performance, fabricate all segments of trunk A from the same reel of cable. Do the same for trunk B (all from the same reel as cable A, or all from a different reel).

Each trunk cable segment must begin and end at bands marked on the cable outer jacket. To minimize signal reflection buildups at tap locations, the trunk cable must have these periodic band markings that identify optimum connector placement.

---

**Testing of fabricated cables**

Any fabricated coaxial cable should be tested as described in the *Universal Control Network Installation* manual before it is installed in the Universal Control Network. If the trunk cable is pulled through conduit or cable tray, it should be tested before and after pulling.

---

**Connectors and tools**

Connectors for RG11-type trunk cable can be ordered from Honeywell as a kit with 24 connectors. The kit also includes a supply of orange and brown cable markers to color-code the ends of cable A and cable B, respectively.

A trunk connector crimping tool is available in a kit from Honeywell. For the part number of the kit, see the appendix of this manual.

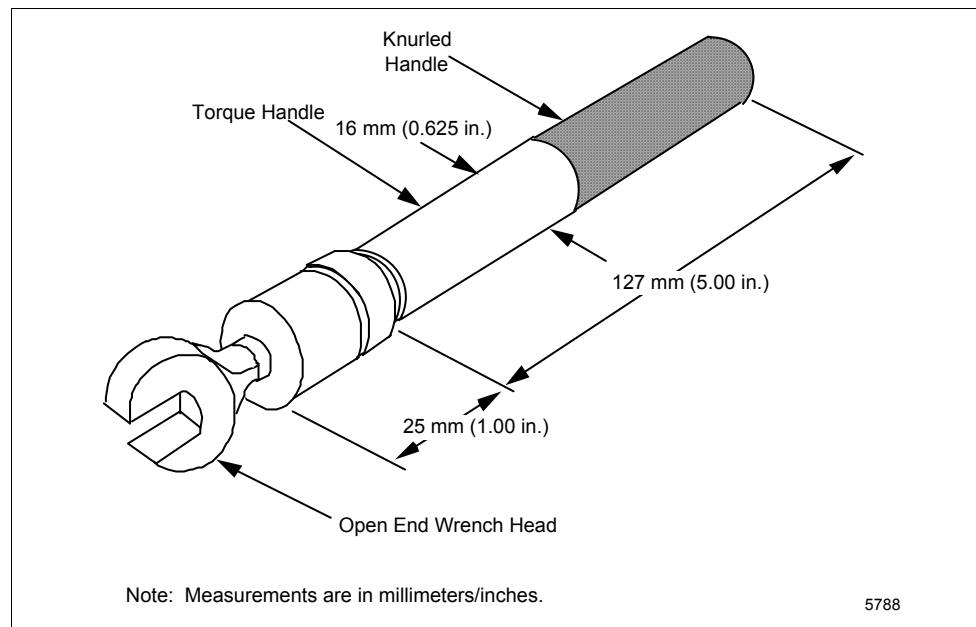
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## 2.10 Connector Torque Wrench

### Wrench description

All cable connectors and 75-ohm terminators must be tightened to 25 inch-pounds with a torque wrench after the connector or terminator has been finger tightened. The suggested tool, illustrated in Figure 2-5, is found in a Honeywell kit, model number MU-NKTQ01, part number 51109612-100. The wrench is factory set to 25 inch-pounds.

Figure 2-5 Connector Torque Wrench



### Wrench sizes

Included in the kit are three hex wrench heads, sizes 11-mm (7/16-inch), 13-mm (1/2-inch), and 16-mm (5/8-inch), to be used with the torque wrench. The heads incorporate a cutout, allowing the wrench to fit over a cable while it is connected.

- The 13-mm (1/2-inch) flare head is used for the drop coax cable connectors, the new style trunk coax connectors, and the new style trunk cable tap terminators.
- The 11-mm (7/16-inch) head accommodates the old style terminators.
- The 16-mm (5/8-inch) head accommodates the old style trunk cable connectors.

### Using the wrench

A click can be felt or heard when the calibrated 25 inch-pounds torque limit has been reached. *Do not overtighten the connector or terminator.*

#### CAUTION

CAUTION—You may not feel the click of the wrench if you place your fingers on the head of the wrench while gripping it. This can cause you to overtighten the connection.

## Section 3 – Trunk Cable Planning

### 3.1 Overview

---

#### Section contents

These are the topics covered in this section:

|      | Topic                                         | See Page |
|------|-----------------------------------------------|----------|
| 3.1  | Overview.....                                 | 17       |
| 3.2  | Trunk Cable Length Versus Allowable Taps..... | 18       |
| 3.3  | Power Line Separation.....                    | 25       |
| 3.4  | Coaxial Cable Installation.....               | 27       |
| 3.5  | Spare Port for Test Purposes.....             | 28       |
| 3.7  | Cable Routing and Trays.....                  | 28       |
| 3.8  | Hazardous Environments.....                   | 29       |
| 3.9  | UCN Grounding.....                            | 29       |
| 3.10 | Cabling Documentation.....                    | 31       |

---

#### Introduction

You must consider a number of parameters in the design of a site Universal Control Network, such as distance, number of UCN taps, routing, and power line proximity. These and other topics are discussed in the following text.

---

#### Follow good design practices in installing a UCN network

Universal Control Network security is enhanced by always connecting both devices of a redundant pair of UCN devices to the same UCN taps, and by connecting redundant UCN devices to taps in the same order along both the A and B trunks.

---

#### **CAUTION**

Never install drop cable in place of trunk cable because drop cable has more attenuation and does not have cable marking bands for proper placement of the connector.

This section provides design parameters and equation which must be followed when designing a Universal Control Network. Be sure you understand and follow them.

---

## 3.2 Trunk Cable Length Versus Allowable Taps

### Introduction

The relationship between the RG11-type trunk cable, the UCN taps, and the RG6-type drop cable lengths is illustrated in Table 3-1 for 2-drop and 4-drop UCN taps. If 8-drop taps are used, see the loading equation later in this section.

The 5-meter (16.4-foot) and 50-meter (164-foot) drop cable lengths are chosen as examples of short and long drop cable lengths.

Table 3-1 Maximum Allowable Taps for RG11-Type Trunk Cable

| Total Cable Length | Drop Cable Length (meters/feet)   |        |                                   |        |
|--------------------|-----------------------------------|--------|-----------------------------------|--------|
|                    | 5/16.4                            | 50/164 | 5/16.4                            | 50/164 |
|                    | Number of 2-Drop UCN Taps Allowed |        | Number of 4-Drop UCN Taps Allowed |        |
| 100/328            | 40                                | 32     | 25                                | 20     |
| 200/656            | 35                                | 27     | 22                                | 17     |
| 300/984            | 30                                | 22     | 19                                | 14     |
| 400/1312           | 25                                | 17     | 16                                | 11     |
| 500/1640           | 20                                | 12     | 13                                | 8      |
| 600/1968           | 15                                | 7      | 10                                | 5      |
| 700/2297           | 10                                | 0      | 7                                 | 0      |
| 800/2625           | 0                                 | 0      | 0                                 | 0      |

Eight-drop UCN taps are not illustrated in the table. Treat the 8-drop UCN taps as two 4-drop UCN taps for a given drop cable length.

### An example configuration

As an example, consider A and B trunk cables with an end to end length of 500 meters (1640 feet). There are four 2-drop taps on each trunk.

As shown in the Table 3-1, a 500-meter (1640-foot) RG11-type trunk cable can accommodate up to twelve 2-drop UCN taps using 50-meter (164-foot) drop cables. For the example given, four 2-drop UCN taps is an acceptable configuration.

### Maximum acceptable trunk length

The maximum acceptable length of a UCN trunk is about 750 meters (2460 feet), depending upon the number of taps.

## 3.2 Trunk Cable Length Versus Allowable Taps, Continued

---

**Advantage of using multi-drop taps**

For a given overall trunk length, 4-drop UCN taps provide more drop connections because they provide twice as many drops as the 2-drop UCN taps without quite doubling the trunk signal attenuation.

For example, assume a 500-meter trunk cable using 50-meter drop cables. Checking Table 3-1 at the intersection of the fifth row (“500/1640”) and the two 50-meter drops columns (“50/164”), we see twelve 2-drop UCN taps or eight 4-drop UCN taps are the maximum allowed.

If you used all 2-drop UCN taps in this situation, you could provide a maximum of 24 drops, while replacing them with 4-drop UCN taps increases the maximum to 32 drops.

---

**Maximum number of taps need shortest cable length**

Table 3-1 lists a minimum trunk cable length of 100-meters (328 feet). For trunk cables less than 100 meters (328 feet), the number of taps allowed is limited by “Rule two” that is discussed below.

---

**Rules for calculating cable lengths**

The rules and equation discussed below can be used for other trunk lengths not listed in Table 3-1, or for networks that employ a mixture of 2-drop, 4-drop, or 8-drop UCN taps, which is the most probable situation in most installations.

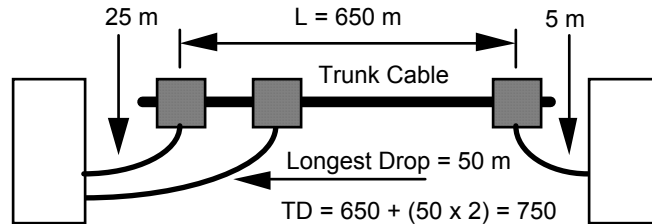
---

## 3.2 Trunk Cable Length Versus Allowable Taps, Continued

### Rule one

**Length compared to Tilt/Distortion Limit**—The first rule is that the total trunk length (L), plus two times the longest drop length must be no greater than the Tilt/Distortion Limit (TD) for the type of trunk cable being used.

For example, the Tilt/Distortion Limit (TD) for standard indoor, outdoor, or armored UCN RG11-type trunk cable is 750-meters (2461-feet).



Then, a 650-meter (2133-foot) trunk cable (L), and at least one 50-meter (164-foot) drop cable, does not exceed the Tilt/Distortion Limit (TD) of 750-meters.

The trunk length (L) you use, must be the longer of the redundant UCN trunk cables, A or B. They may be of equal length, but normally are not because they are routed along different paths to avoid potential damage from the same source. For other Honeywell approved trunk cables, check with your Honeywell sales engineer for their Tilt/Distortion limits.

### Rule two

**Limits imposed by cable bands**—The second rule is that for proper Universal Control Network operation, any two adjacent taps on the trunk can be no closer together than the adjacent band interval markings on the cable's outer jacket, which is 2.6 meters (8.5 feet) for standard UCN RG11-type trunk cable.

In addition, all taps must be placed only at the band intervals; therefore, all RG11-type trunk cables must be integral multiples of 2.6 meters (8.5 feet) in length. For example, in Table 3-1, 400 meters is really 400.4 meters.

### CAUTION

Use only Honeywell-approved cable that is clearly marked with adjacent band intervals. If non-approved cable is used, or taps are not installed at the markings as recommended, UCN performance specifications may not be met.

### Rule three

**Limits imposed by signal attenuation**—The third rule is that the maximum number of cable taps allowed and the length of the trunk is limited by the signal attenuation through the cable taps and cables as follows:

## 3.2 Trunk Cable Length Versus Allowable Taps, Continued

### General equation for computing attenuation

The general equation for networks using a mixture of 2-drop, 4-drop, and 8-drop UCN taps is:

$$(N2' \times 0.3) + (N4' \times 0.5) + (N8' \times 1.0) + (L \times A) + (2 \times D \times M) + (Z \times 0.5) \leq 13$$

where:

| Parameter  | Description                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>N2'</b> | The number of 2-drop UCN taps, excluding any 2-drop UCN taps at either end of the trunk.                                   |
| <b>N4'</b> | The number of 4-drop UCN taps, excluding any 4-drop UCN taps at either end of the trunk.                                   |
| <b>N8'</b> | The number of 8-drop UCN taps, excluding any 8-drop UCN taps at either end of the trunk.                                   |
| <b>L</b>   | The total length of the trunk in meters.                                                                                   |
| <b>A</b>   | The attenuation factor of the trunk cable in dB per meter. That is 0.015 dB/meter for standard or armored RG11-type cable. |
| <b>D</b>   | The longest drop cable used. The maximum is 50 meters (164 ft.).                                                           |
| <b>M</b>   | The attenuation factor of the drop cable in dB per meter. That is 0.025 dB/meter for standard RG6-type drop cable.         |
| <b>Z</b>   | The number of 8-drop taps at the end of a trunk (0, 1, or 2).                                                              |

If a type of drop tap (2, 4, or 8) is not used, set that value (Nn' or Z) to zero.

### Explanation of equation

The basic equation above states that for a mixture of 2-drop, 4-drop, and 8-drop UCN taps, not including the tap at each end of the trunk, the total attenuation along the trunk, through the trunk cable and through all but the end taps, and along the longest tap drop cable must be less than, or equal to, 13 dB. For the 5 Mbps Universal Control Network system, attenuation is specified at the higher signal frequency, 10 MHz.

Note that 2-drop, 4-drop, and 8-drop UCN taps are considered separately in the equation because they have different attenuation values. Also note that D is the length of the longest drop cable in the network.

Remember that N2', N4', or N8' do not include taps at either end of the trunk. You must use the value 0, 1, or 2 for the variable Z to represent the number of 8-drop taps at the ends of the trunk cable.

## 3.2 Trunk Cable Length Versus Allowable Taps, Continued

### Maximum cable length for allowed attenuation

The equation above can be solved for the number of taps allowed for a given length trunk or solved for the maximum trunk length, given the number of required taps. If no 2-drop UCN taps are used, set N2' equal to zero. If no 4-drop UCN taps are used, set N4' equal to zero. If no 8-drop UCN taps are used, set N8' and Z equal to zero.

$$L \text{ (max)}^* = \frac{13 - [(N2' \times 0.3) + (N4' \times 0.5) + (N8' \times 1.0)] - (2 \times D \times 0.025) - (Z \times 0.5)}{A}$$

\* Remember that Rule One and Rule Two on previous pages also apply; therefore, the actual L (max) can be less than that indicated by solving the equation.

Table 3-2 is placed here for reference while solving the equation above for maximum trunk length using the number of required taps from the table. Eight-drop UCN taps are not illustrated in the table.

Table 3-2 Cable Length vs Number of 2-Drop and 4-Drop Cable Taps

| Total Cable Length | Drop Cable Length (meters/feet)   |        |                                   |        |
|--------------------|-----------------------------------|--------|-----------------------------------|--------|
|                    | 5/16.4                            | 50/164 | 5/16.4                            | 50/164 |
|                    | Number of 2-Drop UCN Taps Allowed |        | Number of 4-Drop UCN Taps Allowed |        |
| <b>100/328</b>     | 40                                | 32     | 25                                | 20     |
| <b>200/656</b>     | 35                                | 27     | 22                                | 17     |
| <b>300/984</b>     | 30                                | 22     | 19                                | 14     |
| <b>400/1312</b>    | 25                                | 17     | 16                                | 11     |
| <b>500/1640</b>    | 20                                | 12     | 13                                | 8      |
| <b>600/1968</b>    | 15                                | 7      | 10                                | 5      |
| <b>700/2297</b>    | 10                                | 0      | 7                                 | 0      |
| <b>800/2625</b>    | 0                                 | 0      | 0                                 | 0      |

### Cable Map worksheet

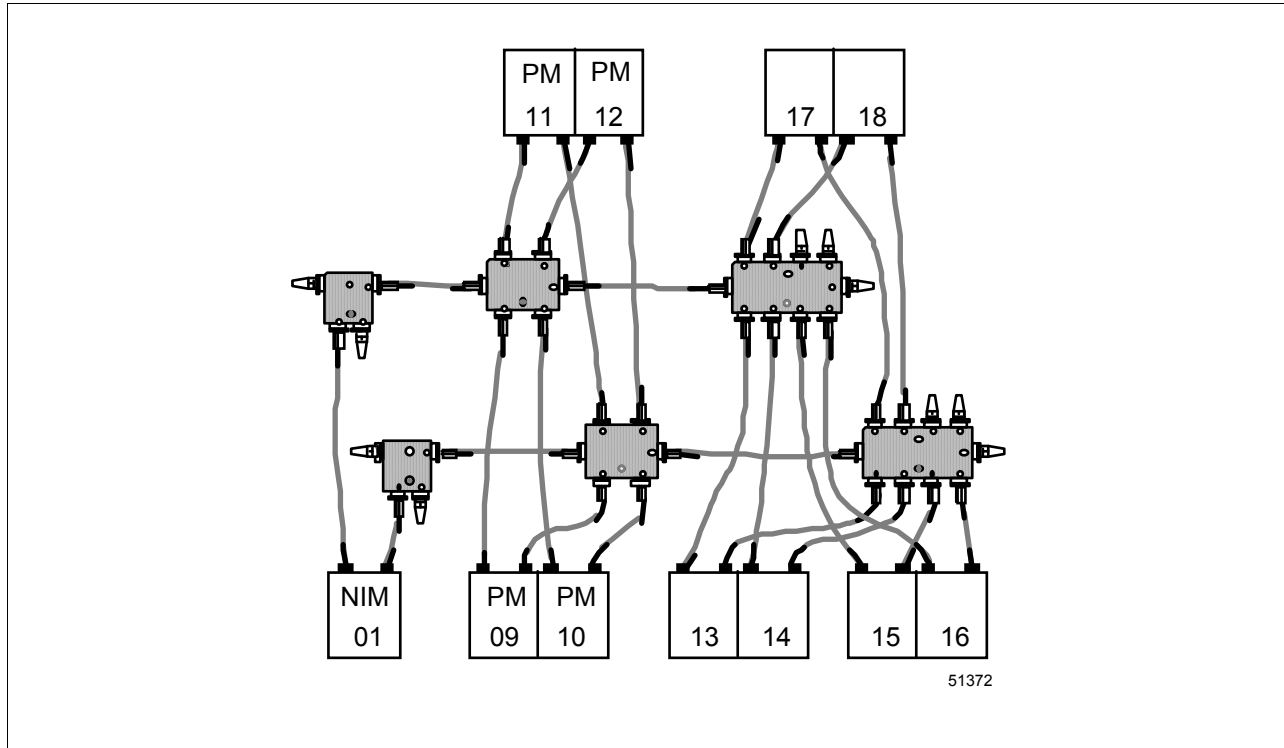
UCN Cable Map worksheets can be used as an aid in planning, and later recording, the topology of your installed network. They are explained in Section 3 and full-sized worksheets are printed in the appendix.

To document your network, a full topology map of your UCN cable system can be drawn using the symbols and information from the worksheets, or multiple copies of the worksheets can be filled out and combined in a loose-leaf notebook or similar binder.

**Network example for determining maximum length**

The following network map in Figure 3-1 shows a 2-drop, 4-drop, and 8-drop cable tap on each trunk. The information contained here is used in the following example for computing the maximum length of the trunk cables and longest tap drop cable to conform with the 13 dB maximum attenuation.

Figure 3-1 Network Topology Example for Computing Maximum Cable Length



**Calculation example  
for maximum length  
of cables**

Assuming the longest drop cable used is 30 meters and using the tap information in Figure 3-1, results in the following parameter values. In the example, the 2-drop and 8-drop taps are at the ends of the cable and not part of N2' and N8', respectively.

|            |   |                                             |
|------------|---|---------------------------------------------|
| <b>N2'</b> | = | 2-drop cable taps per trunk = zero (0)      |
| <b>N4'</b> | = | 4-drop cable taps per trunk = one (1)       |
| <b>N8'</b> | = | 8-drop cable taps per trunk = zero (0)      |
| <b>L</b>   | = | Unknown - Solve equation for length         |
| <b>A</b>   | = | 0.015 dB/meter                              |
| <b>D</b>   | = | Assume drop cable length of 30 meters       |
| <b>M</b>   | = | 0.025 dB/meter                              |
| <b>Z</b>   | = | 1 ea. 8-drop cable tap at end of each trunk |

Equation:

$$L (\text{max}) = \frac{13 - [(N2' \times 0.3) + (N4' \times 0.5) + (N8' \times 1.0)] - (2 \times D \times M) - (Z \times 0.5)}{A}$$

Substituting:

$$L (\text{max}) = \frac{13 - [(0 \times 0.3) + (1 \times 0.5) + (0 \times 1.0)] - (2 \times 30 \times 0.025) - (1 \times 0.5)}{0.015}$$

Factoring, Summing and Dividing:

$$L (\text{max}) = \frac{13 - [(0) + (0.5) + (0)] - (1.5) - (0.5)}{0.015}$$

$$L (\text{max}) = \frac{13 - [0.5] - (2.0)}{0.015}$$

$$L (\text{max}) = \frac{10.5}{0.015}$$

$$L (\text{max}) = 700 \text{ meters}$$

The maximum acceptable length of a UCN trunk is about 700 meters (2296.5 feet), depending upon the number of taps.

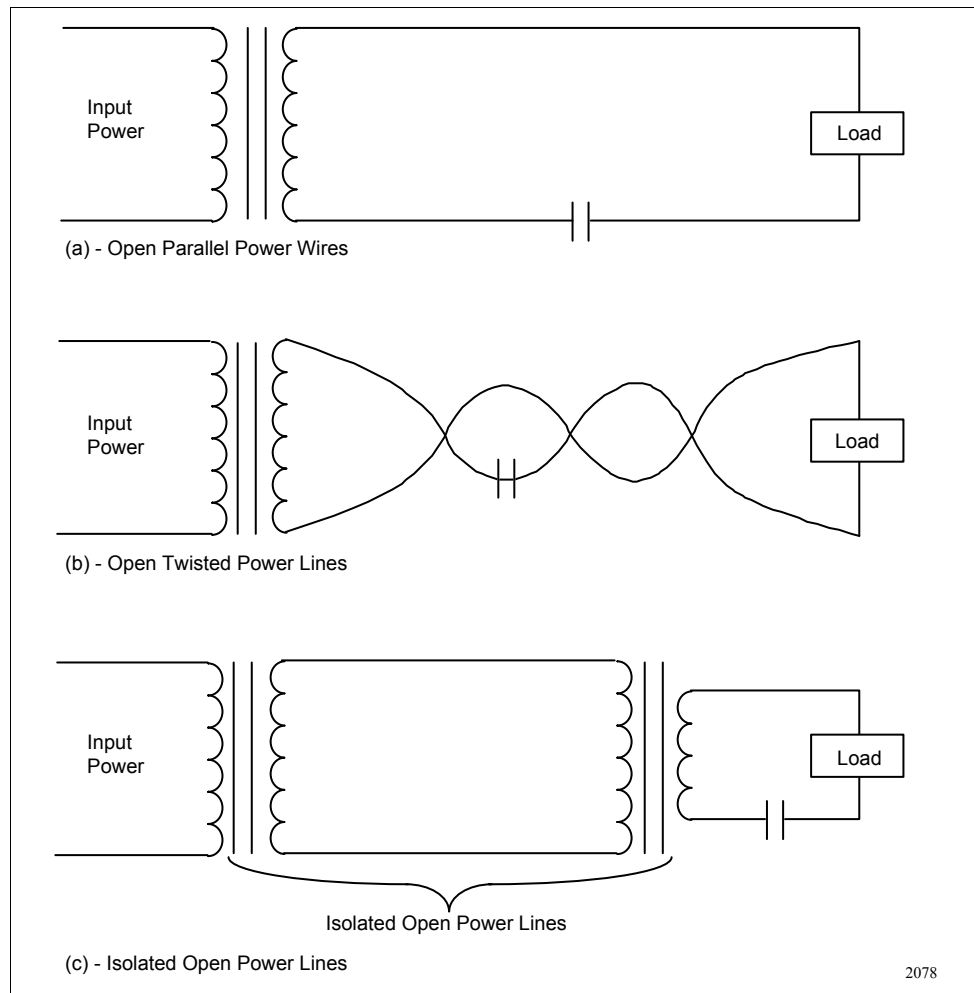
### 3.3 Power Line Separation

#### Introduction

UCN coaxial cable should never be routed through areas of excessive electrical interference. Interference can add or subtract from the ongoing data signals and cause erroneous data transfer.

This section will guide you while routing coaxial cables to reduce the possibility of interference pickup. Refer to Figure 3-2 for power line configurations while using these guidelines.

Figure 3-2 Types of Power Line Configurations



**Source of interference** The most common source of interference is the high frequency magnetic field around power lines. The switching of electrical loads on and off causes a transformer-like coupling of high frequency noise onto the coaxial cables. The problem is extreme when power line voltages are in the kilovolt range.

**Separation from open parallel power line** When loads on open parallel conductor power lines are directly switched, UCN coaxial cables should be separated from the power lines by at least 15 m/kV (50 feet/kV). See Figure 3-2 (a).

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Separation from open twisted power line</b>         | When loads on open but twisted conductor (1 turn/1.5 m or 1 turn/5 feet) power lines are switched directly, UCN coaxial cables should be separated from the power lines by at least 3 m/kV (10 feet/kV). See Figure 3-2 (b).                                                                                                                                                                                                                                                                                                                                                |
| <b>Separation from transformer isolated power line</b> | <p>When power lines have switched loads that are isolated by a transformer, UCN coaxial cables should be separated from the power lines by at least 30 cm/kV (1 foot/kV). See Figure 3-2 (c).</p> <p>This separation requirement is independent of the power lines being parallel, twisted, open, or enclosed in metal.</p>                                                                                                                                                                                                                                                 |
| <b>Separation from enclosed power line</b>             | <p>When power lines are enclosed in metal, eddy currents in the enclosure wall greatly reduce radiated interference. Coaxial cables should be separated from the power line enclosure wall by at least 30 cm/kV (1 foot/kV).</p> <p>This requirement is independent of the power lines being parallel or open. The metal enclosure must have a minimum thickness of 0.76 mm (0.03 inches) for ferrous metal and 2.54 mm (0.1 inches) for nonferrous metal.</p>                                                                                                              |
| <b>Separation from transformers and motors</b>         | The inductance of motor and transformer coils greatly reduces radiated interference. UCN coaxial cables should be separated from the coil's enclosure wall by 30 cm/kV (1 foot/kV).                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Right-angle crossing</b>                            | For right angle crossings between power lines and coaxial cables with a $90 \pm 5^\circ$ perpendicularity, and when the nearest bend takes place past the separation distances given above, all previously discussed separation requirements can be reduced by 3:1.                                                                                                                                                                                                                                                                                                         |
| <b>Coaxial cable shielding</b>                         | Install the coaxial cables in a threaded conduit, or bonded conduit or tray, that is grounded at both ends to AC Safety Ground with a resistance to true earth of one ohm or less. This allows all of the separations that are discussed above to be reduced by a ratio of 3:1. Perpendicular and shielded crossovers correspondingly allow separations to be reduced by a ratio of 9:1. As an example, a 1.2-meter (4-foot) wide, welded wire 2.6-millimeter (10 AWG) mesh, with a 7- or 10-centimeter (3- or 4-inch) opening, can be substituted for the conduit or tray. |

## 3.4 Coaxial Cable Installation

---

### Introduction

Coaxial cable can be installed in a facility in a variety of ways. Use this section as a guide to proper cable installation.

---

### Cable tray

Trays and ladder racks are the most common installation means in industrial facilities. The cable unrolls and pulls onto the rack with a minimum of strain and the full length is open for inspection, repair, or modifications.

Standard UCN indoor trunk cable is made of flame retardant PVC jacket to meet Class II requirements. If used outdoors, this indoor trunk cable must be placed in waterproof conduit to protect it from moisture and ultraviolet rays or optional outdoor RG11-type trunk cable with a polyethylene jacket can be used. Refer to the “Conduits” text below for that type of installation.

---

### **CAUTION**

If the cable is installed in plenums and risers, use plenum-rated cable. See your Honeywell sales engineer for more information about these special cables.

---

### Messenger wire

An appropriate type of coaxial cable can be installed on poles by using a messenger wire. The messenger wire is either molded into the cable jacket or it is taped to the cable while it is being strung.

**PROTECTION FROM LIGHTNING ON POLES**—For lightning protection, place a counterpoise 2 meters (6 feet) above the coaxial cable for the full length of the installation. Bury another counterpoise in the ground below the cable connecting both the counterpoises and the coax messenger wire together at each pole. Ground the counterpoises and the coax messenger wire to ac safety ground at each building entry and at each end. Use 25 mm<sup>2</sup> (4 AWG) wire throughout.

---

### Direct burial

Direct burial of an appropriate trunk cable (normally a cable that is protected with armor and flooding compound) is usually the most economical method of installing cable between buildings when there is no existing tube bank or pipe rack.

Protect the buried cable jacket from puncture with 5 cm (2 inches) of stone-free back fill immediately above and underneath the cable. Ground the armor to ac safety ground at each building entry. For assistance in selecting the cable to be used, contact your Honeywell sales engineer.

---

## Conduits

**Conduits**—Conduits are used for physical protection or for controlling ignition of hazardous process materials. Care is required when installing coax in a conduit. The cable may be damaged by:

- Exceeding its pull strength,
- Exceeding the bend radius limit,
- Tearing the jacket on sharp corners, and
- Immersing in water that accumulates in low areas.

Pitch the conduit by a minimum ratio of 1:100 towards a sump. Limit the distance between junction boxes to reduce the strain on the cable.

---

## 3.5 Spare Port for Test Purposes

### Spare drop ports are useful

Provide a spare port on a trunk cable tap, or a spare trunk cable tap at both ends of the trunk and near the middle of the trunk to permit initial and post installation testing of both trunk cables. Availability of an existing spare drop port enables troubleshooting without adding taps and possibly altering network conditions during troubleshooting. This concept is extremely important for troubleshooting UCNs when system control functions must continue during troubleshooting.

---

### WARNING

*Do not* use the trunk termination at either end of the trunk for testing. Connect test equipment only through a drop cable and tap.

---

## 3.6 Cable Routing and Trays

### Different cable paths should be taken

The Universal Control Network is a redundant cable system that permits independent routing of the A and B cables to maximize network reliability.

---

### LCN and UCN cables

LCN and UCN coaxial trunk cables can be routed in the same conduit or tray, LCN A cables with UCN A cables, and LCN B cables with UCN B cables.

---

### Cables may be combined with field wiring and ground

As discussed under signal compatibility in the planning manuals for the various UCN devices, the coax tray or conduit can be the same one that contains Master Reference Ground, 1-5 V/4-20 mA signals, and shielded cables carrying circuit contact signals of less than 30 V dc/ac peak.

An exception is the 50-conductor FTA cable that is a Class 1 Power Limited source.

---

## 3.7 Hazardous Environments

---

|                             |                                                                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special requirements</b> | <p>The UCN coaxial cable may have to be routed through a process area that handles explosive material, which is typical in oil refineries.</p> <p>Hazardous environment cabling requirements are discussed in Section 4 of this manual.</p> |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## 3.8 UCN Grounding

---

|                                                 |                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ground each trunk cable tap separately</b>   | <p>Separately connect the grounding stud of each trunk cable tap with a 5.0 mm<sup>2</sup> (10 AWG) green or bare conductor (or equivalent thickness flat braid wire) to the nearest ac safety ground.</p>                                                            |
| <div><b>ATTENTION</b></div>                     | <p>Do not daisy-chain tap grounds or use the ground stud to mount the tap.</p>                                                                                                                                                                                        |
| <b>Taps mounted in the cabinet</b>              | <p>When taps that are mounted in a cabinet along with the UCN device(s), the best tap grounding location is the ac Safety Ground bar at the bottom of the cabinet.</p>                                                                                                |
| <b>Taps mounted outside the cabinet</b>         | <p>When taps that are mounted outside a cabinet, the nearest ac Safety Ground, such as the grounded steel frame of a building or a power distribution panel ac Safety Ground, should be used for tap grounding. See Figure 2-1.</p>                                   |
| <b>A single ground point is the best choice</b> | <p>Generally, the best choice for ac Safety Ground for grounding each tap is the one that is closest to the tap and is common for all the taps and UCN devices. As a result, ground currents that are due to the use of different grounding points are minimized.</p> |

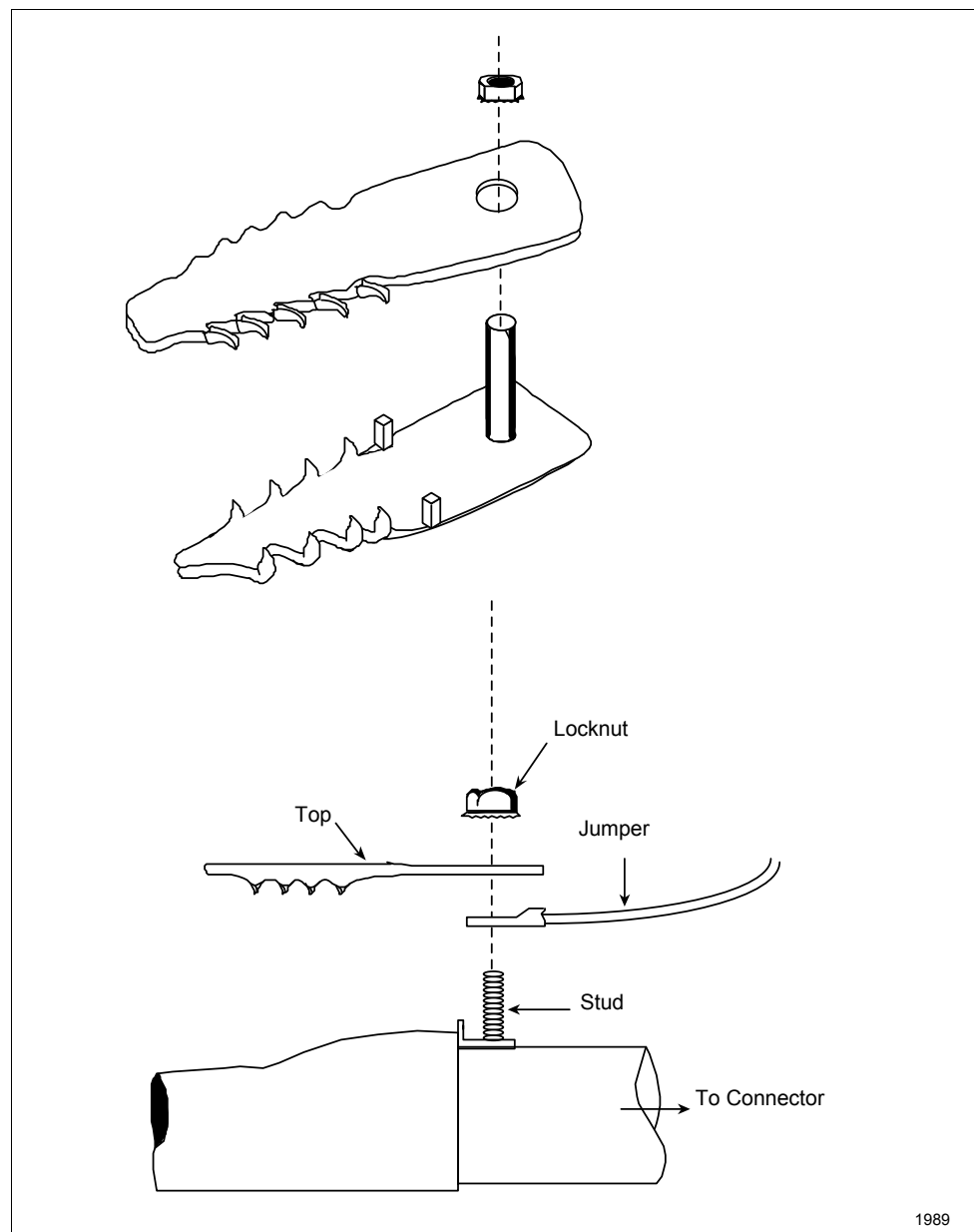
---

## 3.8 UCN Grounding, Continued

### Armored cable grounding

Armored coaxial cable requires grounding the armor for lightning protection and additional noise immunity. Connect the armor at both ends with a 5.0 mm<sup>2</sup> (10 AWG) green or bare conductor (or equivalent thickness flat braid wire) to ac safety ground. Do not ground the armor directly to the UCN taps. It may introduce transients into the coaxial cables. Special clips for the armor connection are often available, as shown in Figure 3-3.

Figure 3-3 Coax Armor Clip



## 3.9 Cabling Documentation

### Document your work

To provide trunk and drop cable connection data for the installer, and also as a permanent record for future reference, we recommend that you develop a detailed network map that identifies each cable segment in the network structure. Use symbols similar to those in Figure 3-4.

### The topology map

The detailed map should include:

- Adherence to the topology rules of the network considering the number of UCN taps versus total trunk length. Review Section 3 in this manual for details.
- Both redundant nodes (two NIMs, two APMs, or two HPMMs) must use the same type and number of A trunk cable and B UCN taps.
- The order of the nodes along the trunk must be the same for both A and B cables.
- Only 75-ohm terminators, not nodes, must be connected at each end of the trunk and every unused trunk cable tap port must be terminated.

### Aids in recording the topology

Figure 3-4 shows worksheets for recording network topology. They are printed in the appendix of this manual. Copy these sheets, fill them out, then transfer that information to a detailed topology map of your network.

Figure 3-4 UCN Cable Network Worksheets

| UNIVERSAL CONTROL NETWORK MAP (WORKSHEET)                                                                                                                                                                                                                                                                        |                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| (Use multiple, consecutive worksheets to show entire network; show terminators at end of trunk)                                                                                                                                                                                                                  |                            |
| Customer _____                                                                                                                                                                                                                                                                                                   | System ID _____ Date _____ |
| Network Name _____                                                                                                                                                                                                                                                                                               |                            |
| UCN Trunk _____ (Enter "A" or "B")                                                                                                                                                                                                                                                                               |                            |
| Tap type: 2 or 4                                                                                                                                                                                                                                                                                                 | Tap type: 2 or 4           |
| Tap ID _____                                                                                                                                                                                                                                                                                                     | Tap ID _____               |
| <p>Diagram showing two network nodes connected by a trunk. Each node has a central box with ports 1, 2, 3, 4, and GND. Port 1 is connected to DROP #1, port 2 to DROP #2, port 3 to DROP #3, and port 4 to DROP #4. A trunk line connects the two nodes, with a legend indicating 'o = ISOLATED TRUNK PORT'.</p> |                            |
| Trunk ID _____                                                                                                                                                                                                                                                                                                   | Trunk ID _____             |
| Length _____                                                                                                                                                                                                                                                                                                     | Length _____               |
| Drop #1 ID/Length _____                                                                                                                                                                                                                                                                                          | Drop #1 ID/Length _____    |
| Drop #2 ID/Length _____                                                                                                                                                                                                                                                                                          | Drop #2 ID/Length _____    |
| Drop #3 ID/Length _____                                                                                                                                                                                                                                                                                          | Drop #3 ID/Length _____    |
| Drop #4 ID/Length _____                                                                                                                                                                                                                                                                                          | Drop #4 ID/Length _____    |

| UNIVERSAL CONTROL NETWORK MAP (WORKSHEET)                                                                                                                                                                                                                                                                                         |                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| (Use multiple, consecutive worksheets to show entire network; show terminators at end of trunk)                                                                                                                                                                                                                                   |                            |
| Customer _____                                                                                                                                                                                                                                                                                                                    | System ID _____ Date _____ |
| Network Name _____                                                                                                                                                                                                                                                                                                                |                            |
| UCN Trunk _____ (Enter "A" or "B")                                                                                                                                                                                                                                                                                                |                            |
| Tap type: 8                                                                                                                                                                                                                                                                                                                       | Tap type: 8                |
| Tap ID _____                                                                                                                                                                                                                                                                                                                      | Tap ID _____               |
| <p>Diagram showing a single network node with a central box and ports 1, 2, 3, 4, 5, 6, 7, 8, and GND. Port 1 is connected to DROP #1, port 2 to DROP #2, port 3 to DROP #3, port 4 to DROP #4, port 5 to DROP #5, port 6 to DROP #6, port 7 to DROP #7, and port 8 to DROP #8. A legend indicates 'o = ISOLATED TRUNK PORT'.</p> |                            |
| Trunk ID _____                                                                                                                                                                                                                                                                                                                    | Trunk ID _____             |
| Length _____                                                                                                                                                                                                                                                                                                                      | Length _____               |
| Drop #1 ID/Length _____                                                                                                                                                                                                                                                                                                           | Drop #5 ID/Length _____    |
| Drop #2 ID/Length _____                                                                                                                                                                                                                                                                                                           | Drop #6 ID/Length _____    |
| Drop #3 ID/Length _____                                                                                                                                                                                                                                                                                                           | Drop #7 ID/Length _____    |
| Drop #4 ID/Length _____                                                                                                                                                                                                                                                                                                           | Drop #8 ID/Length _____    |



## Section 4 – Universal Control Network Extender Planning

### 4.1 Overview

---

#### Section contents

The topics covered in this section are:

|       | Topic                                      | See Page |
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---

## 4.2 Description

---

### Purpose

The Universal Control Network 2-kilometer fiber optic Extender (UCNE) is used to:

- increase the end-to-end distance of a single UCN without resorting to special, wide-diameter coaxial trunk cables.
- increase the UCN signal quality at the ends of the UCN that would otherwise be connected by long coaxial trunk cables.
- implement certain UCN topologies, such as a “star configuration,” that is not possible by using coaxial trunk cables.
- provide ground isolation, when necessary, between UCN trunk segments.

Each UCNE provides coax-to-fiber optic and fiber optic-to-coax cable conversion for the UCN signals.

---

### Implementation

UCNEs are installed in pairs at each end of a redundant fiber optic link; that is, a UCNE-A and UCNE-B at one end of the fiber optic link and a UCNE-A and UCNE-B at the other end of the fiber optic link. Therefore, a total of four UCNEs are required to implement one redundant fiber optic link.

The UCNE, at each end of the fiber optic link, never connects directly to the UCN coaxial trunk, but instead connects to the drop port of a UCN tap on that coaxial segment.

Standard coaxial UCN drop cables (RG6-type) are used to connect the UCNEs to the UCN tap drop ports.

---

## 4.3 Universal Control Network Topologies

### Introduction

There are two classes of UCN topology when using UCN fiber optic Extenders (UCNEs) to link coaxial UCN trunk segments.

- Serial
- Star

### Serial UCN topologies

The simplest “serial” UCN topology consists of two UCN coaxial trunk segments that are interconnected by a single fiber optic link. Figure 4-1 illustrates the “one series link” configuration.

Another serial UCN topology consists of three UCN coaxial trunk segments that are interconnected by two fiber optic links. Figure 4-2 illustrates the “two series link” configuration.

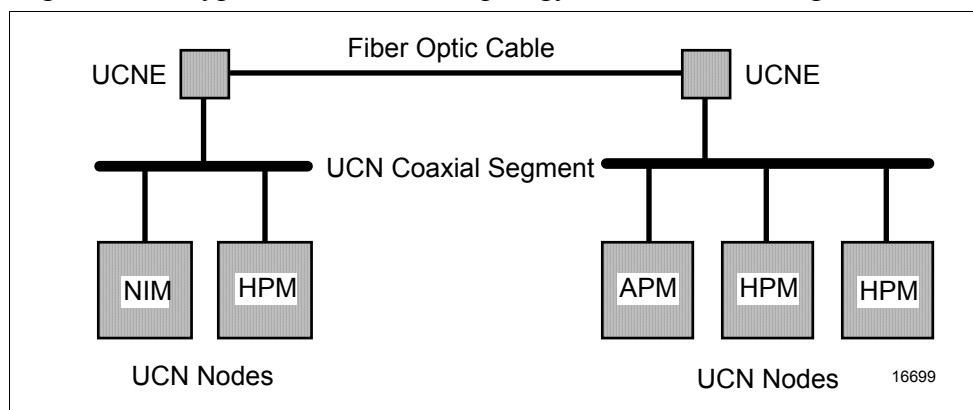
Each UCN redundant fiber optic link requires two UCNEs (UCNE-A and UCNE-B) at each end of the link, or a total of four UCNEs. Two fiber optic links, as illustrated in Figure 4-2, would require eight UCNEs.

The maximum number of series-connected fiber optic links allowed in a physical UCN is two. The use of three or more fiber optic links in a serial topology is prohibited because the delay imposed by the additional UCNE(s) and fiber optic cable could violate the UCN’s slot time setting and cause UCN communication errors.

### Serial UCN topology 2 coaxial segments

Figure 4-1 illustrates a typical serial UCN topology with two UCN coaxial segments and one fiber optic link. Only network “A” is shown. Network “B” is not shown.

Figure 4-1 Typical Serial UCN Topology – Two Coaxial Segments

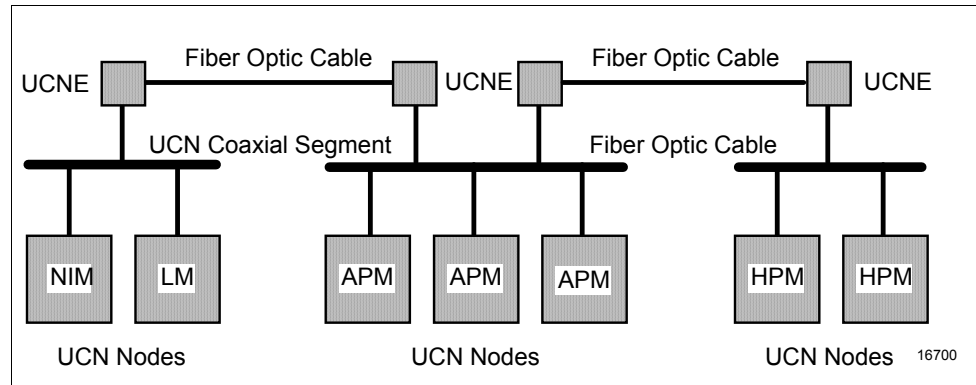


## 4.3 Universal Control Network Topologies, Continued

### Serial UCN topology 3 coaxial segments

Figure 4-2 illustrates a typical serial UCN topology with three UCN coaxial segments and two fiber optic links. Only network “A” is shown. Network “B” is not shown.

Figure 4-2 Typical Serial UCN Topology – Three Coaxial Segments



### Star UCN topology

A “star” UCN topology consists of a central UCN coaxial cable segment with peripheral UCN coaxial cable segments connected to the central UCN coaxial cable segment through UCNEs. Figure 4-3 illustrates a typical star UCN “star” configuration example.

UCNEs must not be used with passive star couplers because passive couplers have excessive optical power loss. Instead, star topology is implemented by using multiple drop ports on a central UCN coaxial segment to branch out through multiple UCN fiber optic links to other UCN coaxial segments. An example of their use is a central control room that contains the NIM(s) with multiple radial UCN fiber optic links (using UCNEs) to remote UCN coaxial segments that accommodate other UCN nodes. Figure 4-3 illustrates four radial fiber optic links to four remote sites.

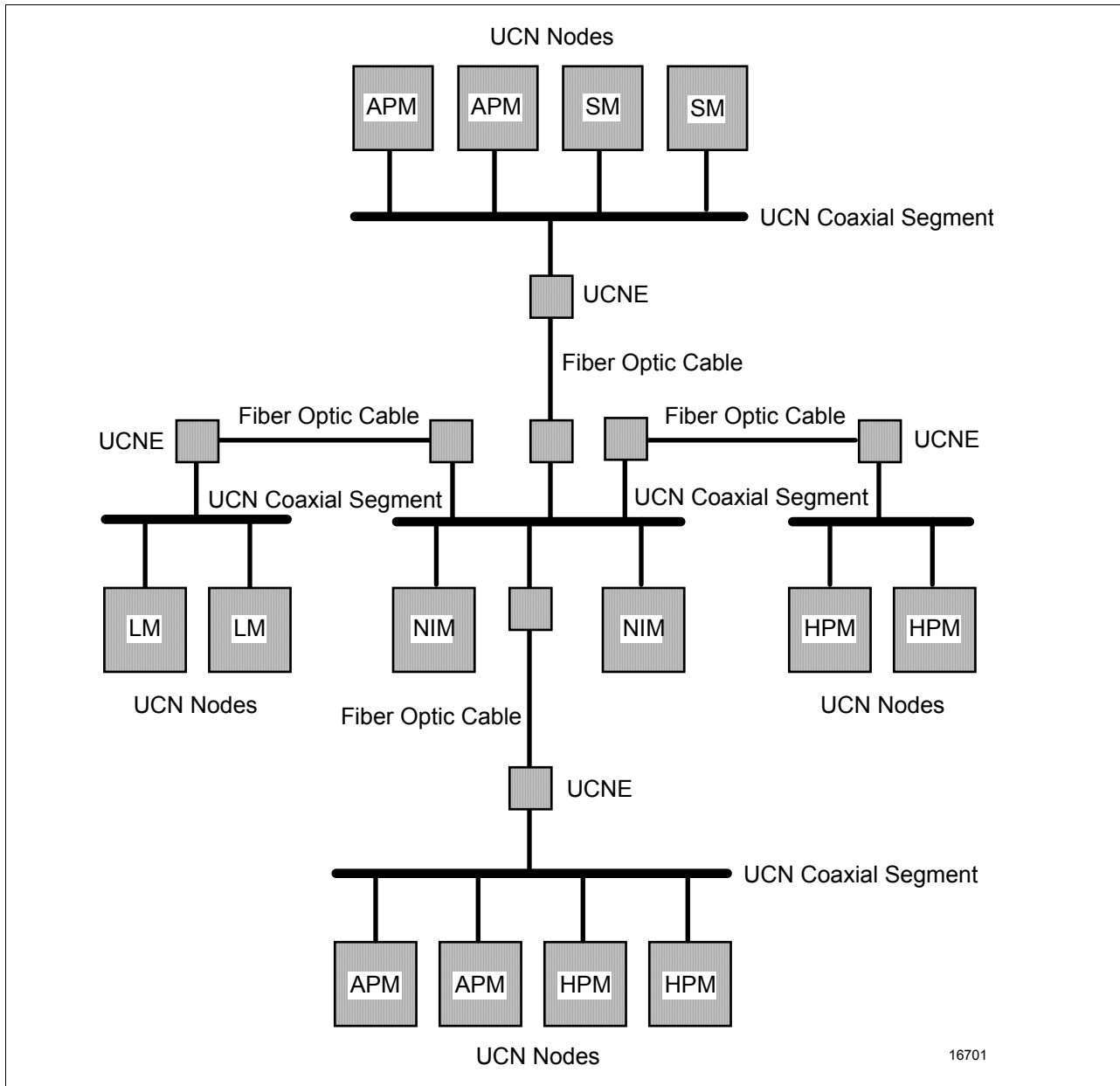
Only one of the redundant UCN cables, A or B, is shown in the figure for simplification.

It is preferable that the NIM(s) is at the center of the “star,” but it is not a requirement, just a recommendation, to minimize the “loss of view” in the unlikely event that a redundant fiber optic link fails completely.

## 4.3 Universal Control Network Topologies, Continued

**Serial UCN topology** Figure 4-3 illustrates a typical star UCN topology with five UCN coaxial segments. Only network “A” is shown. Network “B” is not shown.

Figure 4-3 Typical Star UCN Topology



## 4.4 UCNE Configuration Rules

---

### Configuration rules

Regardless of the UCN topology, the following configuration requirements must be met.

1. The total length of the UCN (network A or B) between any two nodes, including all series-connected coaxial trunk cable segments and fiber optic links, but not including drop cable lengths, must not exceed 2 kilometers. Since any two UCN nodes can potentially communicate with each other, the restriction is determined by the two nodes that are farthest apart from each other along either UCN cable A or UCN cable B and separated by a combination of UCN coaxial trunk cable and UCN fiber optic cable. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time. UCN slot time is the maximum time that any UCN node waits for a response from any other UCN node on the network and is affected by cable and UCNE delays.
  2. Related to rule 1 above, the difference between the total length of UCN network A and UCN network B must be no greater than 500 meters. That is, the UCN A cable and UCN B cable communication paths between the two farthest apart UCN nodes must not be different by more than 500 meters. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time.
  3. Related to rule 1 above, the length of fiber optic cable that connects any two UCNEs must not be longer than 2 kilometers, but there is no minimum length requirement. Violation of this rule could result in UCN communications errors because the optical signal loss budget would be exceeded for the interconnected UCNEs (see the following rule).
  4. Related to rule 3 above, the optical signal loss budget for a fiber optic cable connection between two interconnected UCNEs is 10 dB maximum and must not be exceeded. This includes up to 2 kilometers of fiber optic cable that is specified in Honeywell drawing 51190918 that has a maximum loss of 4 dB/km, two ST-type connectors (one at each end of the fiber optic cable) with a maximum loss of 1 dB per connector, and two cable splices with a maximum loss of 0.5 dB per splice. Violation of this rule could result in UCN communications errors because the optical signal loss budget could be exceeded for the interconnected UCNEs.
  5. The UCN coaxial cable configuration rules found in other sections of the manual also apply for each separate coaxial trunk segment of the network.
-

## 4.5 UCNE Model Numbers

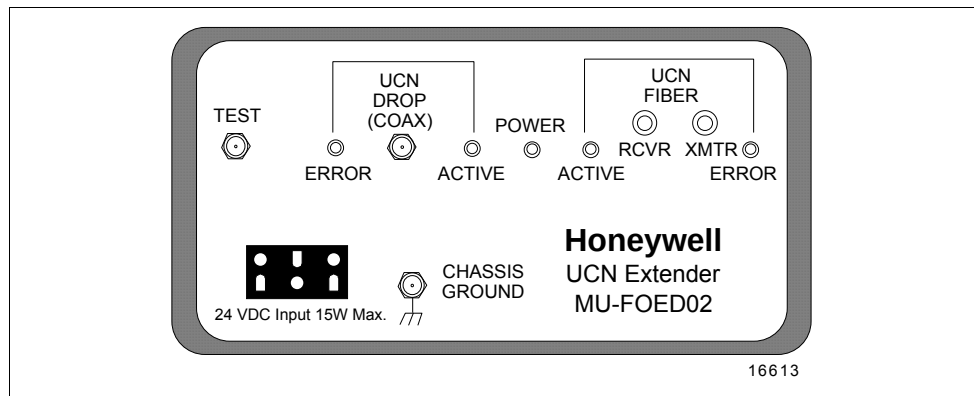
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Model numbers</b>     | <p>The UCNE model numbers that support the maximum two kilometer fiber optic cable distance are:</p> <ul style="list-style-type: none"><li>• MU-FOED02     A pair of 18-36 Vdc powered UCNEs</li><li>• MU-FOEA02     A pair of 85-264 Vac, 47-63 Hz powered UCNEs</li></ul>                                                                                                                                       |
| <b>DC-powered UCNE</b>   | <p>The dc-powered UCNE is used where a PM, APM, or HPM Power System's 24 Vdc is available. Figure 4-4 is an illustration of a dc-powered UCNE.</p>                                                                                                                                                                                                                                                                |
| <b>AC-powered UCNE</b>   | <p>The ac-powered UCNE is typically used at UCN locations where a PM, APM, or HPM Power System's 24 Vdc is not available, such as in cabinets or consoles that house a NIM, LM, or SM. However, the ac-powered UCNE should be powered by UPS-backed ac power if it is expected to continue supporting UCN communications during an ac line power outage. Figure 4-5 is an illustration of an ac-powered UCNE.</p> |
| <b>UCNE independence</b> | <p>UCNEs must be installed independently from each other such that a failed UCNE in a redundant pair (UCNE-A and UCNE-B) can be replaced while the other UCNE continues operating. This is accomplished using the standard mounting kits that are described in subsection 4.6.</p>                                                                                                                                |

## 4.5 UCNE Model Numbers, Continued

### DC-powered UCNE

Figure 4-4 illustrates the front panel of a dc-powered UCNE.

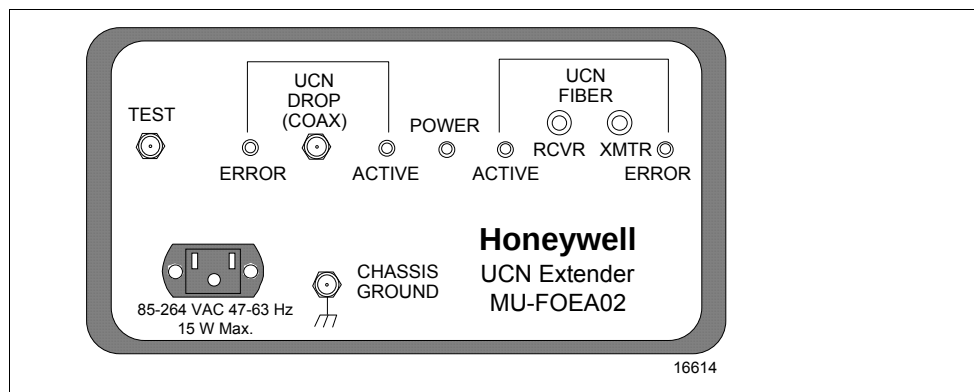
Figure 4-4 DC-Powered UCNE



### AC-powered UCNE

Figure 4-5 illustrates the front panel of an ac-powered UCNE.

Figure 4-5 AC-Powered UCNE



## 4.6 UCNE Mounting Kits

### 4.6.1 PM, APM, and HPM Cabinets



#### SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

---

#### Mounting location

The standard mounting location for a redundant pair of UCNEs is in model MU-FOMK01 mounting kit hardware. The kit's enclosure, which is the same height as an HPM 7-Slot card file but approximately 1 inch wider, can be installed in the Left or Right 7-slot card file position in the card file side of a PM, APM, or HPM cabinet. The mounting brackets will not accommodate two UCNE enclosures. An enclosure can be mounted in either the left or right position on the mounting brackets, but two UCNE enclosures cannot be mounted side by side on the same mounting brackets. Typically, the kit is installed after first mounting all the other PM, APM, or HPM card files that connect to the UCN, and in a cabinet that contains UCN taps because UCNEs connect to the UCN through drop cables to the UCN taps.

#### Mounting kits MU-FOMK01 and MU-FOMK11

The model MU-FOMK01 mounting kit provides for vertical stacking of up to four UCNEs as illustrated by the layout in Figure 4-6. A redundant pair of dc-powered UCNEs (model MU-FOED02) will occupy vertical slots

1 and 3 as illustrated in Figure 4-6. Table 4-1 lists the parts that are included in the model MU-FOMK01 mounting kit.

The model MU-FOMK11 mounting kit provides the hardware to accommodate a second redundant pair of dc-powered UCNEs for use in vertical slots 2 and 4 for an optional second UCN fiber optic link. The second pair of UCNEs must be ordered separately by ordering another model MU-FOED02. Table 4-2 lists the parts that are included in the model MU-FOMK11 mounting kit.

Each of the four vertical mounting slots includes a slide-out mounting tray that attaches to the bottom of a UCNE by using four screws that are provided in the mounting kit.

Mounting positions 1 and 2 are identified by orange-colored "UCN-A" labels, and mounting positions 3 and 4 are identified by brown-colored "UCN-B" labels.

Mounting kits model MU-FOMK01 and MU-FOMK11 include two 5-meter power cables (51201397-002) to provide power to the dc-powered UCNEs. If longer power cables are needed, they must be ordered separately.

**UCN tap drop cables**      A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect a pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the UCNE mounting kits and must be ordered separately.

---

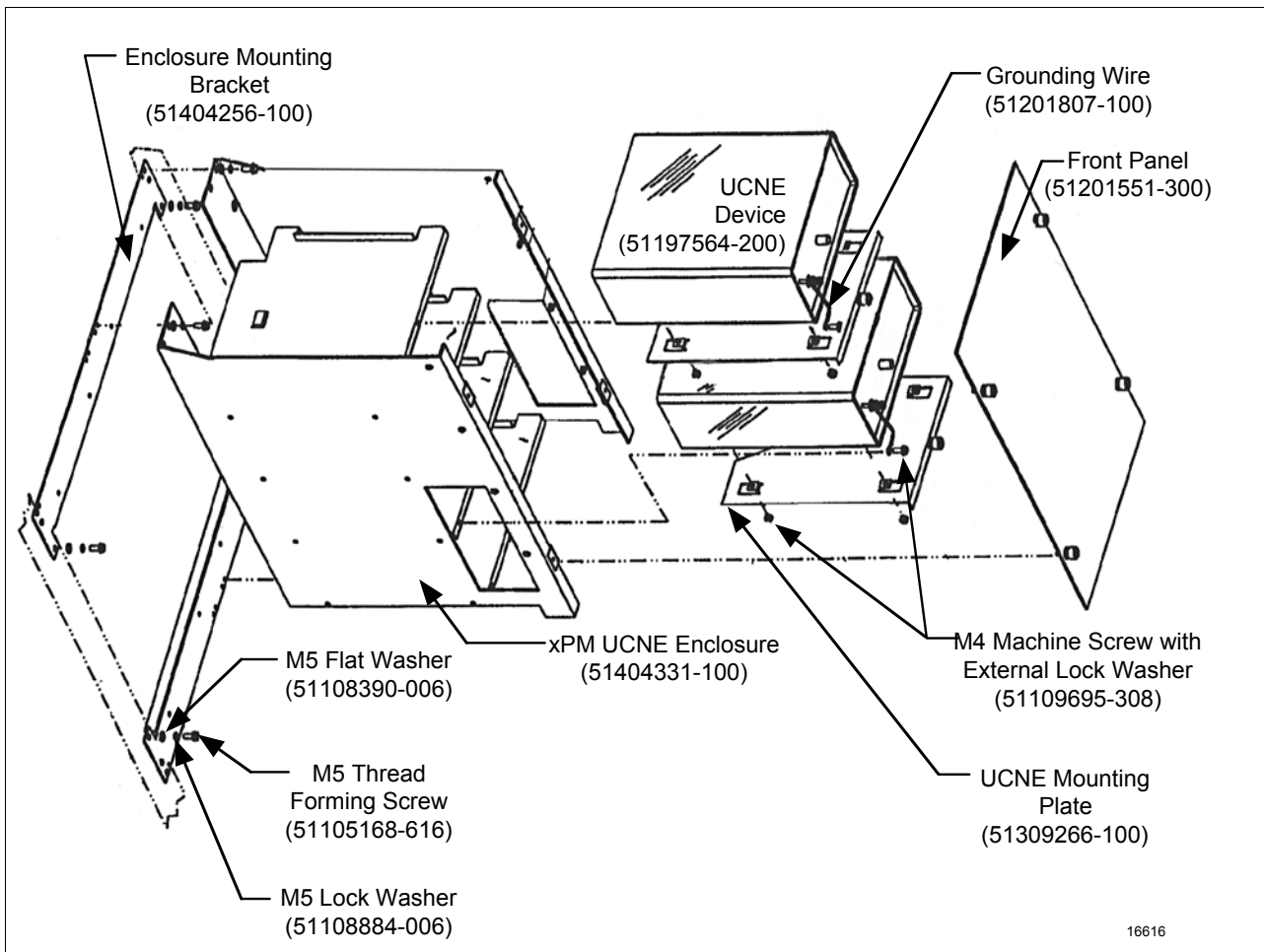
**UCN taps**      If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

|           |                                                  |
|-----------|--------------------------------------------------|
| MU-NTAP02 | Two 2-drop taps with mounting/grounding hardware |
| MU-NTAP04 | Two 4-drop taps with mounting/grounding hardware |
| MU-NTAP08 | Two 8-drop taps with mounting/grounding hardware |

---

**Pictorial view of model MU-FOMK01 kit**      Figure 4-6 is a pictorial view of the model MU-FOMK01 mounting kit. The UCNEs are not included in the kit.

Figure 4-6      Pictorial View of Model MU-FOMK01 Mounting Kit



## 4.6.1 PM, APM, and HPM Cabinets, Continued

### Model MU-FOMK01 mounting kit parts

Table 4-1 is the list of parts in the model MU-FOMK01 mounting kit. The pair of UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOED02.

Table 4-1 Model MU-FOMK01 Parts List

| Description                                   | Part Number  | Quantity |
|-----------------------------------------------|--------------|----------|
| UCNE Enclosure                                | 51404331-100 | 1        |
| Power Cable – 2 meters (6.5 feet)             | 51201397-002 | 2        |
| UCNE Mounting Tray (attaches to UCNE)         | 51309266-100 | 2        |
| Grounding Wire                                | 51201807-100 | 2        |
| Front Panel                                   | 51201551-300 | 1        |
| Enclosure Mounting Bracket (mounts enclosure) | 51404256-100 | 2        |
| Cable Tie                                     | 51190879-001 | 8        |
| M4 Machine Screw                              | 51109695-308 | 10       |
| M5 Thread Forming Screw                       | 51195168-616 | 4        |
| M5 Flat Washer                                | 51108390-006 | 10       |
| M5 Lock Washer                                | 51108884-006 | 10       |

### Model MU-FOMK11 mounting kit parts

Table 4-2 is the list of parts in the model MU-FOMK11 mounting kit. The pair of UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOED02.

Table 4-2 Model MU-FOMK11 Parts List

| Description                       | Part Number  | Quantity |
|-----------------------------------|--------------|----------|
| UCNE Mounting Tray                | 51309266-100 | 2        |
| Power Cable – 2 meters (6.5 feet) | 51201397-002 | 2        |
| M4 Machine Screw                  | 51109695-308 | 10       |
| Grounding Wire                    | 51201807-100 | 2        |

## 4.6.2 LCN Cabinets and Consoles



### SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

### ATTENTION

ATTENTION—The information that follows is supplemental to the PM/APM/HPM cabinet information. Detailed LCN cabinet/console information can be found in the *Local Control Network Planning* and *Local Control Network Installation* manuals.

#### Mounting location

The standard mounting location for a redundant pair of UCNEs in LCN cabinets or consoles is in the UCNE enclosures that are included in the following mounting kits.

|               |                                           |
|---------------|-------------------------------------------|
| MP-FOMK02     | LCN Cabinet 120 Vac UCNE Mounting Kit     |
| MP-FOMK03     | LCN Cabinet 240 Vac UCNE Mounting Kit     |
| MP-FOMK04     | Classic Console 120 Vac UCNE Mounting Kit |
| MP-FOMK05     | Classic Console 240 Vac UCNE Mounting Kit |
| MP-FOMK06-100 | Z-Console Upper Pod UCNE Mounting Kit     |
| MP-FOMK06-200 | Z-Console Lower Pod UCNE Mounting Kit     |

The LCN UCNE enclosures can also be installed in LM and SM cabinets. The model MP-FOMK02 and MP-FOMK03 mounting kits are recommended for that purpose.

#### Model MP-FOMK06 mounting kits

The model MP-FOMK06 mounting kits do not include a pair of replacement leg covers which are required when adding a UCNE enclosure to Z-console furniture. They must be ordered separately. Two factors must be considered before ordering, the height of the console furniture (high, medium, or low) and whether the UCNE enclosure's equipment pod will be the only (upper) pod or a second (lower) pod.

## 4.6.2 LCN Cabinets and Consoles, Continued

---

### Kit descriptions

For each of the kits, a redundant pair of ac-powered UCNEs (model MU-FOEA02), which must be ordered separately, are mounted side by side in a metal enclosure (module). Viewing the rear of the enclosure, the UCNE on the right side of the enclosure is connected to the UCN-A cable and the UCNE on the left side of the enclosure is connected to the UCN-B cable.

Labels on the enclosure identify the UCN-A and UCN-B mounting locations. The labels are color-coded orange and brown, respectively, to agree with the standard UCN redundant cable color-coding scheme.

Each mounting kit includes two ac power cables to power the pair of ac-powered UCNEs that must be installed in the enclosure. If redundant ac power sources are available, it is recommended that each UCNE in a redundant pair be connected to a different ac power source to maintain operation of at least one UCNE if a power source should fail. An Uninterruptible Power System (UPS) is recommended to maintain ac power to the UCNEs during ac line outages.

---

### UCN tap drop cables

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect the pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the LCN UCNE mounting kit and must be ordered separately.

---

### UCN taps

If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

|           |                                                  |
|-----------|--------------------------------------------------|
| MU-NTAP02 | Two 2-drop taps with mounting/grounding hardware |
| MU-NTAP04 | Two 4-drop taps with mounting/grounding hardware |
| MU-NTAP08 | Two 8-drop taps with mounting/grounding hardware |

---

## 4.7 Grounding

---

**Proper grounding required**

The chassis ground stud at the front of the UCNE must be connected to Safety Ground, similar to the grounding of a UCN tap.

A #10 AWG ground wire is provided in all the UCNE mounting kits to connect the UCNE's front panel ground stud to ground on the metal enclosure in which the UCNEs are installed.

The eventual connection to Safety Ground depends on proper grounding of the metal enclosure. As required when installing a UCN tap, the installer must confirm that there is electrical continuity between the UCNE's front panel grounding stud and the cabinet Safety Ground bus bar after the installation is complete. The Safety Ground bus bar is typically located at the bottom of the cabinet.

---

## 4.8 Fiber Optic Cable

---

**Recommended cable**

The multi-node fiber optic cables recommended for use with a UCNE are the same as the 62.5-micron fiber optic cables that are recommended for the Honeywell I/O Link Extender in the PM, APM, or HPM. Honeywell drawings 51190918 (indoor) and 51190919 (outdoor) define the cables. The cable has 4 dB/km maximum optical attenuation according to the signal loss budget requirements of subsection 4.4.

Other fiber optic cable types, such as the 100-micron fiber optic cable, have not been tested with the UCNE by Honeywell and are not recommended. The UCNE is not compatible with single-mode fiber optic cable.

---

**Proper cable construction**

UCN indoor fiber optic cable assemblies should be fabricated using the information in Honeywell drawing 51195498. The RCVR and XMTR fibers at each end of the cable should be labeled as such to agree with the connector labeling on the UCNEs.

---

**Cable composition**

A UCN fiber optic cable link consists of two transmit-receive pairs of optic fibers (UCN-A and UCN-B). If possible, cables that have high installation costs, such as cables routed in conduit or buried in the ground should have spare optic fibers.

---

## 4.9 Fiber Optic Cable Routing

---

|                                 |                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Routing methods</b>          | Fiber optic cable can be routed underground by using direct burial cable, through the air by using outdoor aerial cable, or in cable or electrical wiring trays. Fiber optic cable is immune to interference from electromagnetic fields or transmissions. Fiber optic cable is safe to route through Intrinsic Safety (IS) areas with no danger of explosion. |
| <b>Cable A and B separation</b> | Subject to the configuration rules in subsection 4.4, the A and B fiber optic links should take different routes to a common destination to avoid simultaneous damage to both cables.                                                                                                                                                                          |

---

### 4.9.1 Direct Burial

---

|                                    |                                                                                                                                                                                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Under ground burial hazards</b> | Heavy duty cables can be buried directly in the ground. Care should be taken to minimize the effects of freezing water, the heaving of rocks caused by the ground freezing, ground disruption because of construction, and rodents.                                                                 |
| <b>Type of cable to use</b>        | The most durable jacket material for environmental hazards, such as crushing and abrasion, is polyethylene. An inner layer of steel armor provides additional protection from rodents. Use of heavy duty cables can sometimes be avoided by protecting the cable with conduit or polyethylene pipe. |

---

### 4.9.2 Aerial Lashing

---

|                               |                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aerial lashing methods</b> | Aerial Lashing methods are similar to those used for electrical cables. Most cables are compatible with helical lashing, clamping, or tied mounting. |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

---

### 4.9.3 Vertical Installations

---

**Cable clamping**

Cable in vertical trays, raceways, shafts, etc., should be secured at frequent intervals so that cable weight is supported evenly. Clamping intervals can be as short as one meter (3.3 feet) outdoors to prevent wind slapping and minimize ice loading, or as long as 15 meters (49.2 feet) indoors. When securing the fiber optic cable, care must be taken not to compress the optic fibers or exceed the minimum bend radius specifications.

---

**Fiber migration and cable loops**

Optic fiber in vertical installations does not break because of its own weight. However, for vertical runs of 15 meters (50 feet) or greater, excess optic fiber can migrate downward. The crowding of excess optic fiber at the bottom can cause an undesirable increase in signal attenuation.

This downward migration of optic fiber in vertical runs can be reduced by placing loops in the cable approximately 0.3 to 0.5 meter (1 to 1.5 feet) in diameter at the top, bottom, and at 15-meter (50-foot) intervals.

---

### 4.9.4 Indoor Requirements

---

**Cable jacket building code restrictions**

Building code requirements frequently do not allow cables with polyethylene jackets. Jackets of polyvinyl chloride are frequently restricted to conduits, while fluoropolymer or other approved jacket material is required for use in cable trays and air plenums. Cable with suitable jacket material must be selected for the application.

---

### 4.9.5 Loose Buffered Cable

---

**Ambient temperature variations**

Where ambient temperature variations are 20°C or greater on a daily basis, the life of the optic fibers can be significantly reduced. In applications where the ambient temperature is not controlled, loose buffered cable must be specified.

---

**Fire code requirements**

Loose buffered cable is available in polyethylene jacket material only. Fire codes may dictate that the indoor portion of the cable installation be in metal conduit, or if the temperature is controlled, a splice may be required to convert to a tight-buffered cable with a PVC or fluoride copolymer jacket material.

## 4.9.6 Number of Optic Fibers

---

|                                         |                                                                                                                                                                                                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Multi-optic fiber cable required</b> | Two optic fibers are required for each UCN-A or UCN-B link, one for transmit and one for receive.                                                                                                                                                              |
| <b>Cable damage possible</b>            | As insurance against future damage, such as optic fiber breakage, or encountering excessive loss in any one optic fiber, the inclusion of spare optic fibers should be considered. This is especially important for cables that have a high installation cost. |

---

## 4.9.7 Cable Installation

---

|                                    |                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use professional installers</b> | It is recommended that the fiber optic cable be installed by professional fiber optic cable installers. The installers will provide the cable, install the cable, attach ST-type connectors, and test the connection with an Optical Time Domain Reflectometer (OTDR) and a power meter.                  |
| <b>62.5 micron cables</b>          | <p>62.5 micron cable usage is controlled by Honeywell drawing 51190918 for indoor cable and Honeywell drawing 51190919 for outdoor cable.</p> <p>Indoor cables are available with 2, 4, or 6 optic fibers.</p> <p>Outdoor aerial and direct burial cables are available with 4, 6, or 8 optic fibers.</p> |
| <b>100 micron cables</b>           | 100 micron cable usage cannot be recommended at this time for the UCNE because it has not been tested with the UCNE.                                                                                                                                                                                      |

---

## 4.10 Indoor Cable Bend Radius

### Cable bend requirements

Fiber optic cable is easily damaged by excessive bending or kinking of the cable, so a minimum bend radius is established for each different cable size and cable construction. Indoor cable is the most flexible and outdoor direct burial cable is the least flexible. Table 4-3 specifies the minimum bend radius for indoor cable.

In the table, short term minimum bend radius refers to the minimum bend radius that is safe during installation. Long term minimum bend radius refers to the minimum bend radius that is safe after the cable installation is completed and settled.

Table 4-3 Minimum Bend Radius for Indoor Cable

| Fibers Per Cable | Minimum Bend Radius (Short Term) | Minimum Bend Radius (Long Term) |
|------------------|----------------------------------|---------------------------------|
| 2                | 17.0 cm (6.7 in.)                | 13.0 cm (5.1 in.)               |
| 4                | 22.0 cm (8.7 in.)                | 16.5 cm (6.5 in.)               |
| 6                | 26.0 cm (10.2 in.)               | 20.0 cm (7.9 in.)               |
| 8                | 30.0 cm (11.8 in.)               | 23.0 cm (9.1 in.)               |

## 4.11 Cable Construction

### Fiber optic cable selection

The selection of fiber optic cable is dependent upon satisfying installation and environmental requirements without exceeding the maximum optical signal losses. The factors to be considered when selecting the cable are:

- the total fiber optic cable losses.
- the cable requirements caused by the desired routing. Routing requirements can include direct burial, conduits, trays, raceways, plenums, etc.
- construction code requirements.

### Installation precautions

The installation procedures for placement of fiber optic cables are the same as for electrical wires. Care should be taken to avoid yanking, flipping, or warping that can result in the unnecessary tightening of the optic fibers. Fiber optic cables should not be subject to foot traffic or crushing forces. Sharp bending and scraping at entrances and covers should be avoided.

## 4.12 Cable Splices and Connections

---

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Planning the cabling</b>                | <p>When planning a system installation, design the system to have the minimum number of splices. For example, convert outdoor cable to indoor cable when entering a cabinet where the bend radius of outdoor cable will not fit in the cabinet. Where this splicing must be made, sufficient cable length must be provided for a splice loop. Thirty to 45 centimeters (12 to 18 inches) on each cable end is the usual allowance for a service loop. Also, when entering the equipment cabinet, sufficient cable length must be allowed for breakout (stripping and fanning out) and termination of the individual optic fibers.</p> |
| <b>Cable splice protection</b>             | <p>Splices cannot withstand tensile forces and must be housed in a strain relief assembly. Moisture entry into the splice can cause degradation of performance. Therefore, the splice enclosure must be sealed, and if necessary, the splice encapsulated to minimize moisture entry.</p>                                                                                                                                                                                                                                                                                                                                             |
| <b>Cable breakout</b>                      | <p>Breakout eliminates the outer sheath, leaving the more flexible individual optic fibers for routing within the cabinet. Tight buffered indoor cables (discussed in subsection 4.9.5) provide strength members with each optic fiber which eliminates the need to use breakout kits to add strength members for each optic fiber.</p>                                                                                                                                                                                                                                                                                               |
| <b>Use of a breakout kit</b>               | <p>In outdoor loose buffered, gel-filled cable designs, there is no strength member or protective jacket for the individual optic fibers. Termination of this type of cable requires the use of a breakout or fan-out kit. The kit provides strength members with flexible jackets that are placed over the optic fibers after the outer jacket is removed. These kits can be installed over optic fibers as much as 10 meters (33 feet) in length. The kit provides optic fiber protection for in-cabinet routing and termination.</p>                                                                                               |
| <b>Cables with connectors preinstalled</b> | <p>If the cable length is accurately determined and the cable does not have to be pulled through a conduit, the cables can be ordered from a fiber optic cable assembly supplier with connectors installed on the ends, pre-tested, and with protective covers on the ends.</p>                                                                                                                                                                                                                                                                                                                                                       |

---

## 4.13 Signal Loss Budget

### Introduction

The optic fiber cable/connector/splice signal loss budget is the maximum allowable optic signal loss between a transmitting UCNE and its receiving UCNE partner when interconnected by fiber optic cable that includes ST-type connectors and optional splices. It is the difference between the minimum transmit power and the minimum receiver sensitivity, minus the desired signal loss margin (3 dB) that is necessary to account for the decreasing transmit power characteristics of aging transmitters.

### ATTENTION

In the calculations that follow, the loss in the ST-type connector at the cable end that attaches to the UCNE front panel XMTR connector is already included in the transmitter output specification and does not need to be included in the cable/connector/splice signal loss budget. However, any other connectors or splices must be included, including the ST-type connector at the far end of the fiber link that attaches to the UCNE partner's RCVR connector.

### Signal loss budget calculation

The signal loss budget for the UCNE is calculated as follows. Table 4-4 provides a conversion table between dBm and microwatts.

Transmit Power - Receiver Sensitivity = Budgeted Signal Loss

| <u>Component</u>                      | <u>dB Level</u>  |
|---------------------------------------|------------------|
| Transmitter                           | -14.0            |
| Receiver                              | <u>(-) -27.0</u> |
|                                       | 13.0             |
| Less transmitter power loss over time | <u>(-) 3.0</u>   |
| Signal loss budget                    | 10.0             |

Summarizing the above calculation, the rated transmit power is -14 dB prior to transmitter aging. The rated receiver sensitivity is -27 dB. The transmitter aging loss is 3 dBm maximum. The worst case signal loss budget for the cable, splice, and connector loss is 10 dB.

## 4.13 Signal Loss Budget, Continued

### Power level conversion chart

Table 4-4 is useful for converting optical power levels between two common units of measurement, dBm and microwatts, for various types of optical power meters

Table 4-4 dBm to Microwatts Conversion Table

| <b>dBm</b> | <b>Microwatts</b> |
|------------|-------------------|
| +3         | 2000              |
| 0          | 1000              |
| -3         | 500               |
| -6         | 250               |
| -9         | 125               |
| -12        | 62                |
| -14        | 40                |
| -15        | 32                |
| -17        | 20                |

### Types of splices

Two types of splices are available, mechanical and fusion.

The mechanical-type of splice consists of a glass tube into which the optic fiber is secured by ultra violet (UV) curable epoxy.

The fusion-type splice is produced by carefully aligning the optic fibers and then fusing them together by heating the optic fibers.

In both cases, the signal loss in the resulting splice is very dependent upon the skill of the technician. The losses at all splices and cable installations need to be certified and recorded at the time of installation for future reference as mentioned in subsection 4.14.

## 4.13 Signal Loss Budget, Continued

### Cable span distance formula

As discussed previously in subsection 4.4, UCNE Configuration Rules, the maximum total length of UCN-A or UCN-B, including fiber optic links and coaxial trunk cables, must not exceed 2 kilometers which is based on cable delays. Therefore, because of cable/connector/splice attenuation of optical signals in a fiber optic link, it is possible that the fiber optic link distance may be less, but never more than 2 kilometers.

To calculate the maximum distance of the fiber optic link based on attenuation, use the following equation.

$$\frac{10.0 - (\text{losses in splices}) - (\text{losses in ST-type connectors})}{\text{Maximum cable loss in dB/km}} = D$$

Where D is the distance in kilometers.

The maximum distance for 62.5/125 micron cable with two mechanical splices and an ST-type connector at each end of the cable (one connector is already included in the loss signal budget previously discussed) can be calculated using Tables 4-5 and 4-6.

$$\frac{10.0 - (4.0 \times 0.5) - (1 \times 1.0)}{4} = 1.75 \text{ kilometers}$$

Table 4-5 Optical Power Loss

| Type of Splice    | Average Loss (dB) | Maximum Loss (dB) |
|-------------------|-------------------|-------------------|
| Fusion            | 0.2               | 0.3               |
| Mechanical        | 0.3               | 0.5               |
| ST-Type Connector | 0.5               | 1.0               |

Table 4-6 Optic Fiber Losses (@ 820 nm UCNE Wavelength)

| Type of Fiber   | Average Loss (dB/km) | Maximum Loss (dB/km) |
|-----------------|----------------------|----------------------|
| 62.5/125 Micron | 3.75                 | 4.0                  |

## 4.14 Power Level Measurement

### **Record post installation measurements**

---

After the cables have been connected to the fiber optic transmitters and power has been applied to the equipment, measurements of transmitter output, cable loss and power input to the fiber optic receivers should be recorded for future reference. This procedure is detailed in the *Universal Control Network Installation* manual.

---



## Section 5 – UCN Intrinsic Safety

### 5.1 Overview

---

#### Section contents

These are the topics covered in this section:

| Topic |                                                   | See Page |
|-------|---------------------------------------------------|----------|
| 5.1   | Overview.....                                     | 57       |
| 5.2   | Hazardous Area Classifications .....              | 58       |
| 5.3   | UCN Cabling in Division 2 (Zone 2) Areas.....     | 59       |
| 5.4   | UCN Cabling in Division 1 (Zones 0, 1) Areas..... | 60       |
| 6.1   | Overview.....                                     | 61       |
| 5.2   | Cable Routing.....                                | 62       |

---

#### Introduction

This section details safety requirements for Universal Control Networks located in areas where ignitable or explosive materials may be present.

---

#### Why intrinsic safety?

Certain processes use ignitable or explosive materials. Local electrical codes require that electrical devices that are located in, or are connected to, such process areas have some type of control to prevent accidental ignition of the process material.

---

#### What should be considered?

In this section, intrinsic safety terminology for discussing such UCN installations is defined. Next are the specific intrinsic safety requirements for UCN cables in such installations.

For information about intrinsic safety applications in UCN network devices, see the appropriate planning manuals for those devices or consult with your Honeywell sales engineer.

---

## 5.2 Hazardous Area Classifications

### Applicable electrical codes

Hazardous materials are classified by a variety of terms. The terminology for two commonly used electrical codes is summarized in Table 5-1. These are the National Electrical Code (NEC) in the USA and the standards of the International Electrotechnical Commission (IEC) in Switzerland. The table is not complete. Check your own electrical codes for more information.

Table 5-1 Hazardous Location Classifications

| NEC          | IEC          | Environment                                                                                                                                          |
|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Class I      |              | Explosive gases or vapors are present.                                                                                                               |
| Class II     |              | Combustible dusts are present.                                                                                                                       |
| Class III    |              | Ignitable fibers or flyings are present.                                                                                                             |
| Division 1   | Zone 0       | A location where a hazardous concentration of gases or vapors exist approximately 10-100 % of the time (subject to interpretation of classes above). |
| Division 1   | Zone 1       | A location where a hazardous concentration of gases or vapors exist approximately 1-10 % of the time (subject to interpretation of classes above).   |
| Division 2   | Zone 2       | A location where a hazardous concentration of gases or vapors exist approximately 0.1-1 % of the time (subject to interpretation of classes above).  |
| Nonhazardous | Nonhazardous | Hazardous vapors exist less than 0.1% of the time.                                                                                                   |
| Group A      | Group IIC    | A hazardous atmosphere containing acetylene or other similar gases or vapors.                                                                        |
| Group B      | Group IIC    | A hazardous atmosphere containing hydrogen or other similar gases or vapors.                                                                         |
| Group C      | Group IIB    | A hazardous atmosphere containing ethylene or other similar gases or vapors.                                                                         |
| Group D      | Group IIA    | A hazardous atmosphere containing pentane or other similar gases or vapors.                                                                          |
| Group E      |              | A hazardous atmosphere containing metal dust, such as aluminum.                                                                                      |
| Group F      |              | A hazardous atmosphere containing carbon black, coal, or coke dust.                                                                                  |

When installing a system, the application of this table depends on whether the process is classified as hazardous or nonhazardous, the designated class and group, and the division of the electrical installation and material handling areas.

Note that the area where the UCN cabling is located does not necessarily have the same category as the material handling area where the process is located or the control area where the cable drivers are located.

### An Example

For example, in consulting this table, oil refiners use hazardous materials that are defined by the NEC as belonging to Class I, Groups B, C, and D. In this case, the complete installation has to be appropriately planned.

The divisions (zones) are shown to be subject to interpretation. For example, an area that has a hazardous atmosphere for 1 hour in 10,000 hours (0.01 % of the time) is far off the chart for hazardous areas. As such, it seems to be a safe and nonhazardous area to install electronic equipment.

However, when the equipment has a *hot component* that is capable of causing ignition 100% of the time, explosion is imminent with any process emission. This is a poor location to install nonapproved (for hazardous areas) electronics and the area should be classified as requiring Division 2 (Zone 2) compliance, regardless of Table 5-1.

---

## 5.3 UCN Cabling in Division 2 (Zone 2) Areas

---

### Division 2 certified

For certain electrical codes, UCN cabling to and from the PM, APM, HPM, LM, or SM can be placed in Division 2 (Zone 2) areas, and the wiring techniques and materials can be those approved for use in ordinary (nonhazardous) locations.

---

### Standard UCN cabling approved for Division 2

UCN coaxial cable has a maximum normal operating signal level of 66 dBmV (2 Vac rms); therefore, the normal signal level is not enough to cause ignition from the opening, shorting, or grounding of the coaxial cable circuit.

As such, the standard coaxial cables usually can be routed through a Division 2 area on open ladder racks.

---

## 5.4 UCN Cabling in Division 1 (Zones 0, 1) Areas

---

### Choices for Division 1 protection

UCN coaxial cables ordinarily cannot cause ignition of hazardous material if broken; however, the coaxial cable drivers do not have any special feature for limiting energy during a circuit breakdown.

When permitted by local electrical codes, cabling in Division 1 (Zones 0, 1) areas most economically use one or more of the following techniques.

- Zener barriers,
- Fiber optic isolation,
- Routing under the hazardous area (burial),
- Routing around the hazardous area, or
- Routing through explosion proof conduit.

---

### Achieving Division 1 protection for UCN cabling

Zener barrier and fiber optic isolation techniques are not available for UCN coaxial cabling.

The alternative then, if it is not possible to go around or go underground past a Division 1 (Zones 0, 1) area, is to install the UCN coax in explosion proof conduit.

---

### Explosion proofing

This technique involves enclosing the electrical device (PM, APM, HPM, etc.) and conductors throughout the Division 1 (Zones 0, 1) hazardous area in housings designed to contain the explosion.

This has been the traditional method of connecting electrical devices to a hazardous process.

---

## Section 6 – Coaxial Cable Routing Considerations

### 6.1 Overview

---

#### Section contents

These are the topics covered in this section:

| Topic |                                         | See Page |
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| 6.1   | Overview.....                           | 61       |
| 6.2   | Cable Routing.....                      | 62       |
| 6.3   | External Installation of UCN Taps ..... | 63       |
| 6.4   | Internal Installation of UCN Taps ..... | 65       |

---

#### Introduction

This section discusses the optional methods available when planning the routing and installation of the Universal Control Network cables and supporting components.

---

#### Network overview

The Universal Control Network (UCN) is a network of Network Interface Modules (NIMs), process control network devices, coaxial trunk cables, UCN taps, and coaxial drop cables that interconnect the network. The NIM interfaces the Local Control Network (LCN) to the UCN.

There are many different methods of routing the cables and locating the UCN taps and the drop cables. The guidelines for accomplishing this task were discussed in previous sections in this manual.

The suggested methods discussed in this section should aid in the planning of the Universal Control Network installation and enhance the integrity of the network.

---

## 6.2 Cable Routing

### Advantages of routing trunk cables separately

The redundant UCN trunk cables, cable A and cable B, should be routed through different areas. The areas should be selected based on the probability that if one trunk cable is accidentally damaged, the other trunk cable will likely remain undamaged and operational.

Other than this restriction, there are many options. The cables can be installed in conduit, strung through the air with messenger wire support, placed in cable troughs, routed under a raised floor that supports the equipment, buried in the ground, etc.

### Location of UCN taps

By considering the combination of trunk and drop cable lengths, internal or external trunk cable tap placement can be chosen to satisfy your site plan.

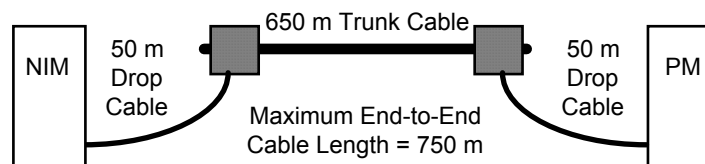
UCN taps located inside the equipment cabinet may provide more protection for the tap, whereas trunk cables installed externally from the equipment enclosures above false ceilings or under false floors may offer more protection for both the trunk and tap.

### Length of trunk cable

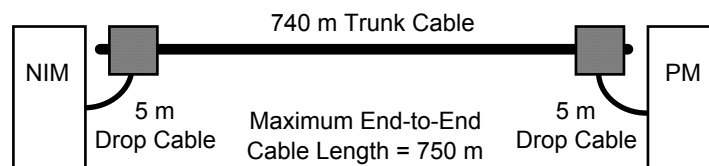
The total length of the trunk cable is limited by the type of cable used and the length of the longest drop cable. The maximum length can be determined by the formula discussed in subsection 3.2 in this manual.

### A method of increasing the effective length of a UCN cable system

For discussion purposes, consider a 650 meter (2133 foot) trunk cable with two network devices consisting of a NIM at one end of the trunk and a PM at the other. Fifty meter (164 foot) drop cables are at each end. The system, diagrammed below, meets the maximum allowable criteria of 750 meters (2461 feet) end-to-end for standard UCN coaxial cables.



If you cannot reach across the physical distance between the NIM and the PM, consider designing a trunk and drop combination shown that below.



The trunk cable is now 740 meters (2428 feet) long and the drop cables at each end of the trunk are 5 meters (16.4 feet) long. In this case, the physical distance between the network devices is greatly increased, but the maximum end-to-end cable length requirement is not broken.

## 6.3 External Installation of UCN Taps

### **Taps must be mounted securely**

When UCN taps are installed external a UCN equipment cabinet, they must be permanently and securely mounted to a grounded fixed structure so that the cables can be strain relieved to maintain a controlled, minimum bend radius that protects the characteristics of the cable. See Figure 6-1 and Figure 6-2.

Figure 6-1 Installation of a Trunk Cable Tap above a False Ceiling

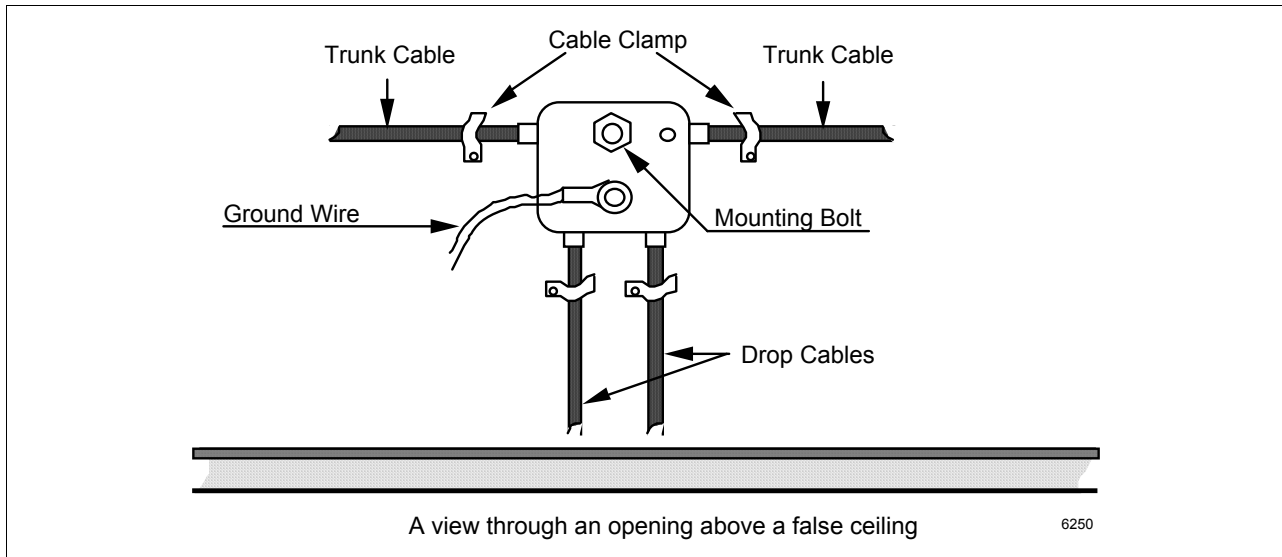
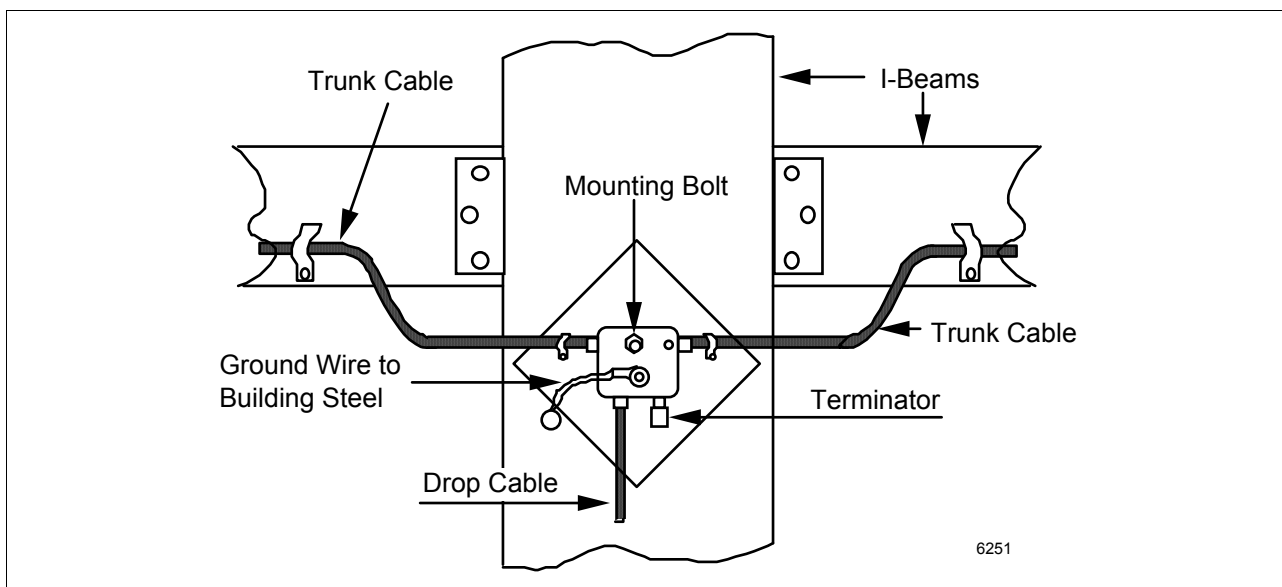


Figure 6-2 Installation of a Trunk Cable Tap on an I-Beam



|                                                         |                                                                                                                                                                                                  |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Protect taps from physical damage</b>                | Taps must be protected so that they are not abused by bumping or by exposure to the elements such as water and sunlight.                                                                         |
| <b>Allow for routing drop cables to network devices</b> | When the UCN taps are installed external to an enclosure containing the network device, the drop cables must be routed into the enclosure. That cable length cannot exceed 50 meters (164 feet). |
| <b>Allow for redundant cable routes</b>                 | Separate paths for the redundant A and B drop cables should be considered if they are in a hazardous location where they can be damaged.                                                         |

## 6.4 Internal Installation of UCN Taps

### Drop taps can be mounted in TPS cabinets

There are various methods of installing UCN taps in a network device enclosure, such as an Advanced Process Manager cabinet or the cabinet containing a Network Interface Module (NIM). Special tap mounting brackets are available in a kit that includes the UCN taps.

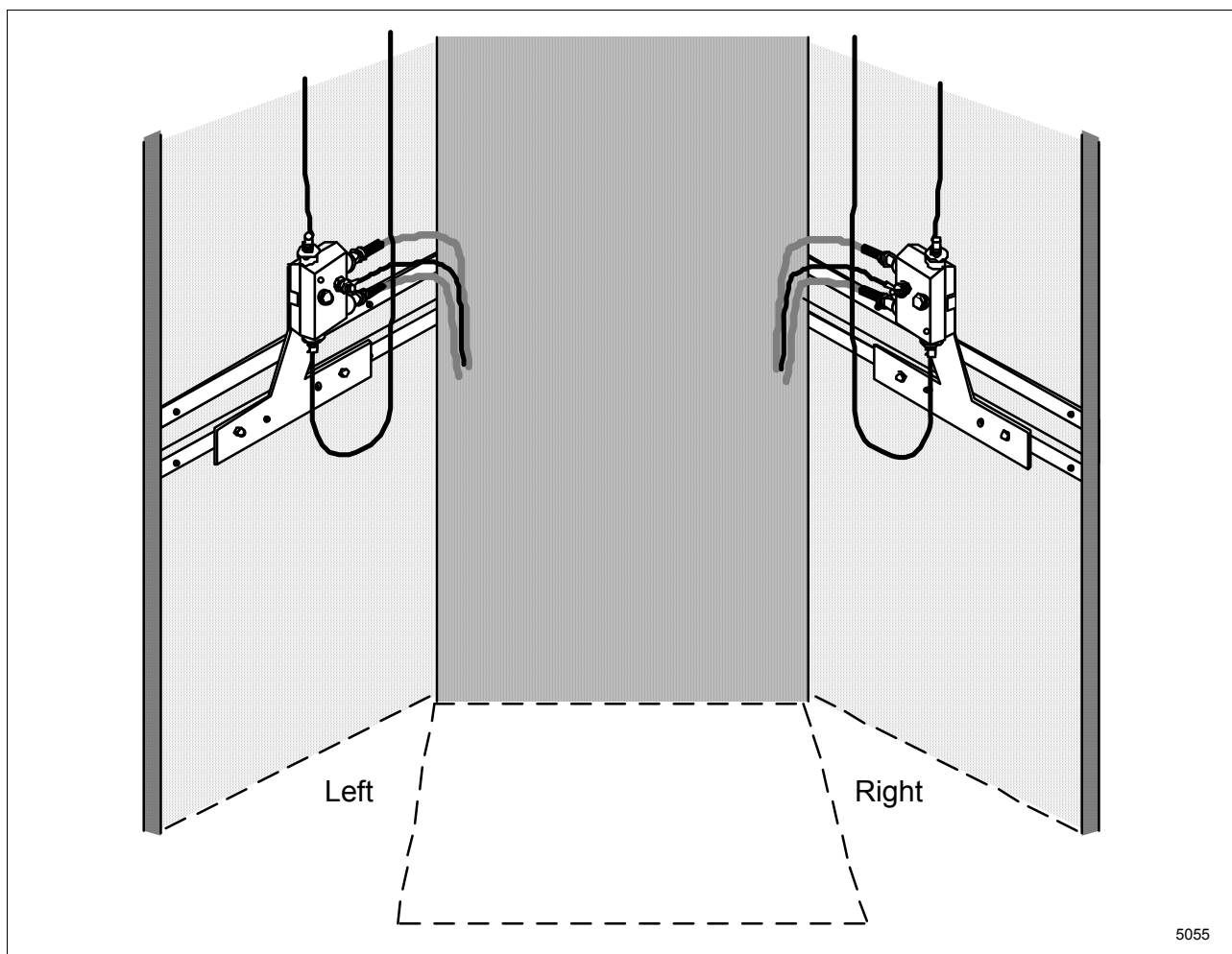
### Preinstalled taps and drop cables

The cabinets containing the network device also can be ordered with the taps and drop cables already installed.

### UCN taps installed on sides of PM/APM/HPM cabinet

Figure 6-3 shows the installation of the UCN taps on the brackets attached to the center cross members at the right and left sides of the Process Manager, Advanced Process Manager, or High-Performance Process Manager cabinet. UCN trunk cable A connects to the left side trunk cable tap and trunk cable B connects to the right side trunk cable tap.

Figure 6-3 Installation of UCN Taps in a PM/APM/HPM Cabinet



## 6.4 Internal Installation of UCN Taps, Continued

### Cabinet mounting details

The center cross members of the Markhon and Rittal cabinets are not the same, requiring different mounting methods. Figure 6-4 and Figure 6-5 show the same mounting bracket with the trunk cable tap installed in the Markhon and Rittal type cabinets, respectively. The 2-drop taps are illustrated, but the brackets also accept the 4-drop and 8-drop taps.

Installation instructions can be found in the installation manuals for the Process Manager, Advanced Process Manager, or High-Performance Process Manager.

Figure 6-4 Trunk Cable Tap Mounting in a Markhon Cabinet

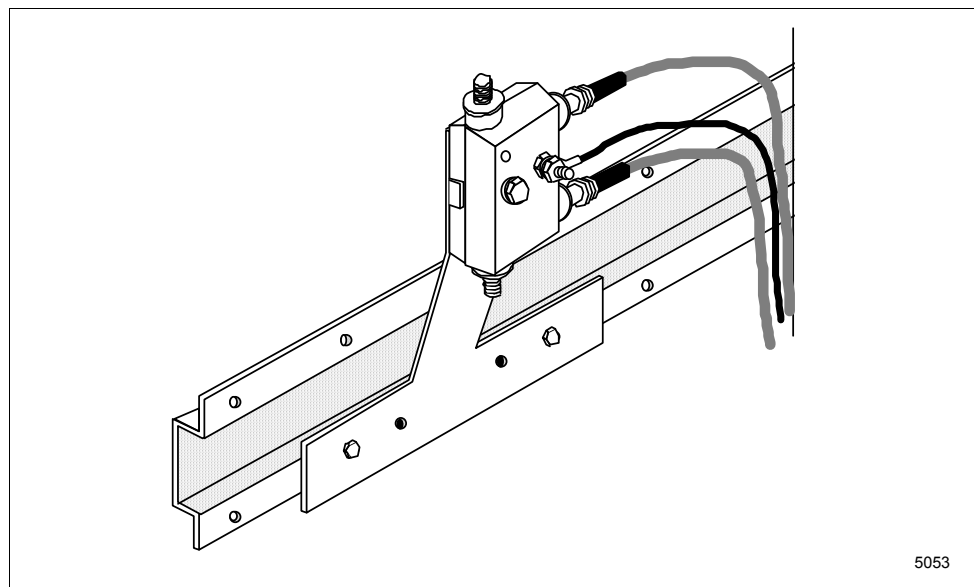
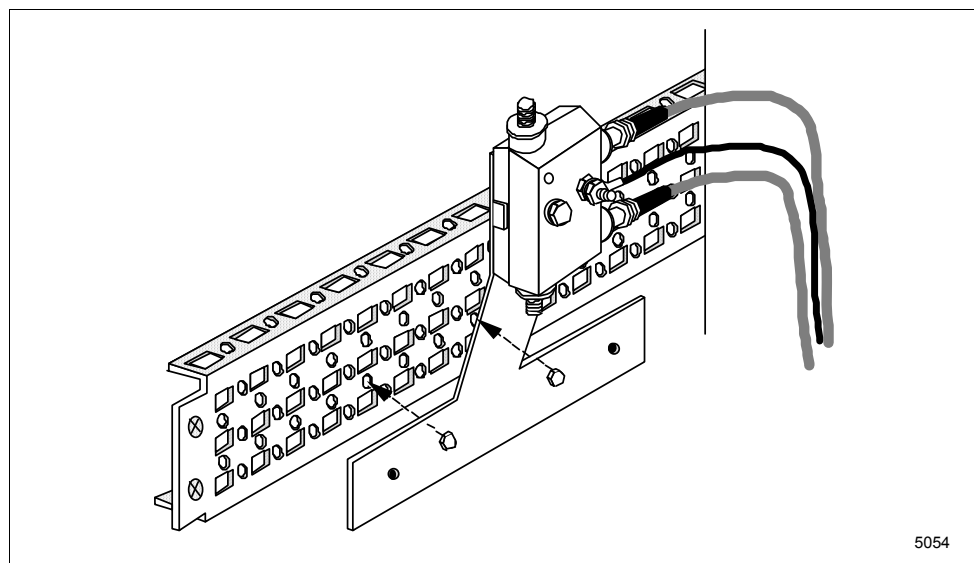


Figure 6-5 Trunk Cable Tap Mounting in a Rittal Cabinet



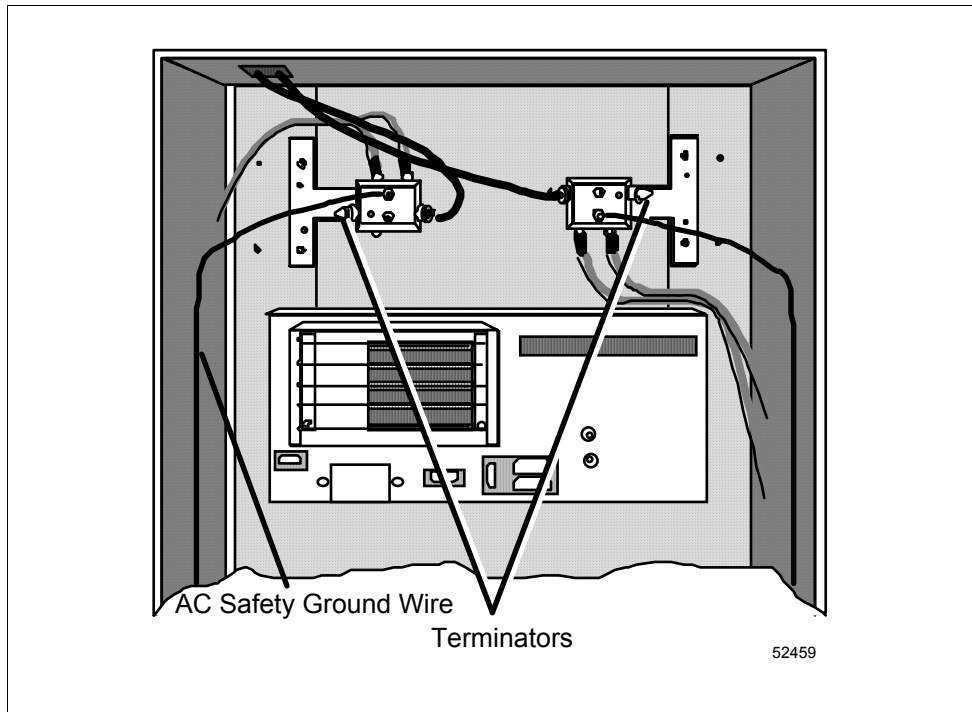
## 6.4 Internal Installation of UCN Taps, Continued

### UCN taps installed on back of an LCN cabinet

Figure 6-6 illustrates the location of the trunk cable tap mounting brackets in the Local Control Network (LCN) cabinet where Network Interface Module (NIM) would normally be installed.

Figure 6-6 (and Figure 2-1) illustrates the NIM connected to one end of the UCN. While this is a possible configuration, it is not necessary that the NIM be installed at the end of the Universal Control Network.

Figure 6-6 UCN Trunk Cable Tap Mounting in an LCN Cabinet



### Tap termination and grounding

The taps at each end of the trunk cables must be terminated with a 75-ohm terminator as illustrated in Figure 6-6. note also that the taps are grounded to AC Safety Ground.

### Additional information

See the *Universal Control Network Installation* manual for additional information.



## Section 7 – UCN Parts List

### 7.1 Overview

---

**Section contents**

These are the topics covered in this section:

| Topic |                 | See Page |
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| 7.2   | Parts List..... | 70       |

---

**Introduction**

Various lengths of RG6-type UCN drop cable and RG11-type UCN trunk cable is available from Honeywell when installing a Universal Control Network. The connectors, taps, 75-ohm terminators, and tools required to install the cables are also available from Honeywell.

## 7.2 Parts List

### UCN Parts List

Part numbers for UCN-related components are listed in Table 7-1.

Table 7-1 UCN Spare Parts

| Name                         | Description                                                                | Part Number  |
|------------------------------|----------------------------------------------------------------------------|--------------|
| UCN Drop Cable               | RG6 UCN drop cable (pair) 0.5 meter                                        | 51195153-900 |
|                              | 1.5 meter                                                                  | 51195153-901 |
|                              | 2.0 meters                                                                 | 51195153-002 |
|                              | 2.5 meters                                                                 | 51195153-902 |
|                              | 3.5 meters                                                                 | 51195153-903 |
|                              | 4.5 meters                                                                 | 51195153-904 |
|                              | 5.0 meters                                                                 | 51195153-005 |
|                              | 10 meters                                                                  | 51195153-010 |
|                              | 20 meters                                                                  | 51195153-020 |
|                              | 30 meters                                                                  | 51195153-030 |
|                              | 40 meters                                                                  | 51195153-040 |
|                              | 50 meters                                                                  | 51195153-050 |
| UCN Trunk Cable              | RG11 UCN trunk cable (pair) – internal application                         | 51195199-xxx |
|                              | 2.6 meters                                                                 | 51195199-002 |
|                              | 5.2 meters                                                                 | 51195199-005 |
|                              | 10.4 meters                                                                | 51195199-010 |
|                              | 20.8 meters                                                                | 51195199-020 |
|                              | 31.2 meters                                                                | 51195199-030 |
|                              | 52.0 meters                                                                | 51195199-050 |
|                              | 101.4 meters                                                               | 51195199-100 |
|                              | 200.2 meters                                                               | 51195199-200 |
|                              | 400.4 meters                                                               | 51195199-400 |
| UCN Trunk Cable              | UCN trunk cable RG11 – external application                                |              |
|                              | xxx meters (762 maximum)                                                   | 51191607-xxx |
|                              | 152.4 meters                                                               | 51191607-152 |
|                              | 304.8 meters                                                               | 51191607-305 |
|                              | 762.0 meters                                                               | 51191607-762 |
| UCN Trunk Cable              | UCN trunk cable RG11 – internal application                                | 51190899-xxx |
| UCN Trunk Tap                | 2-Drop tap assembly kit (pair)                                             | 51201394-100 |
|                              | 4-Drop tap assembly kit (pair)                                             | 51201395-100 |
|                              | 8-Drop tap assembly kit (pair)                                             | 51109485-100 |
| UCN Tap Terminator           | UCN tap terminator                                                         | 51190782-100 |
| Trunk Splice Kit             | Splice kit for RG11 cable                                                  | 51195160-100 |
| Trunk Cable Connector Kits   | RG6 trunk cable connector kit without tool                                 | 51195161-100 |
|                              | RG11 trunk cable connector kit without tool                                | 51195162-100 |
| Cable Connector Tools        | RG6 drop cable connector tool                                              | 51190797-200 |
|                              | RG11 trunk cable connector tool                                            | 51190795-200 |
| Connector Torque Wrench Tool | 25 inch/pound torque wrench kit with three wrench heads for UCN connectors | 51109612-100 |

**EUCN Parts List**      Part numbers for EUCN-related components are listed in Table 7-1a.

Table 7-1a      EUCN Spare Parts

| <b>Name</b>                   | <b>Description</b>                                | <b>Part Number</b> |
|-------------------------------|---------------------------------------------------|--------------------|
| Honeywell Control Firewall 9  | Honeywell Control Firewall 9 (8 ports + 1 uplink) | 51405047-175       |
| Cisco IE 3000 Ethernet switch | Industrial 8-Port Ethernet switch                 | NE-SW508S          |
| Copper expansion module       | 8-port copper expansion module                    | NE-SW58TM          |
| Fiber expansion module        | 8-port fiber expansion module\                    | NE-SW58FM          |



## Section 8 - Enhanced Universal Control Network

### Section contents

These are the topics covered in this section:

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| 8.2 | Understanding EUCN network topology ..... | 74       |
| 8.3 | Hardware Components .....                 | 76       |

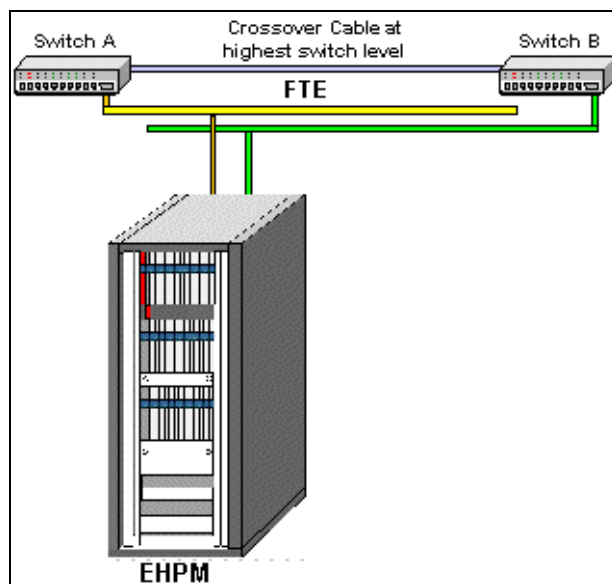
### 8.1 Fault Tolerant Ethernet (FTE)

#### Introduction

Fault Tolerant Ethernet (FTE) on EUCN is dedicated to the control mission of providing fault tolerance, quick response time, and the security required for industrial control applications.

FTE is a single network topology with redundancy. This redundancy is achieved using Honeywell's software and commercially available components such as IE3000 Cisco switches. The FTE Network contains redundant networking components such as switches, cabling, and redundant network interface adapters. The Honeywell software and the FTE-enabled components allow network communication to occur over an alternate route when the primary route fails. The highest-level FTE network switches are inter-connected using a crossover cable.

Figure 8-1 FTE Network Connection

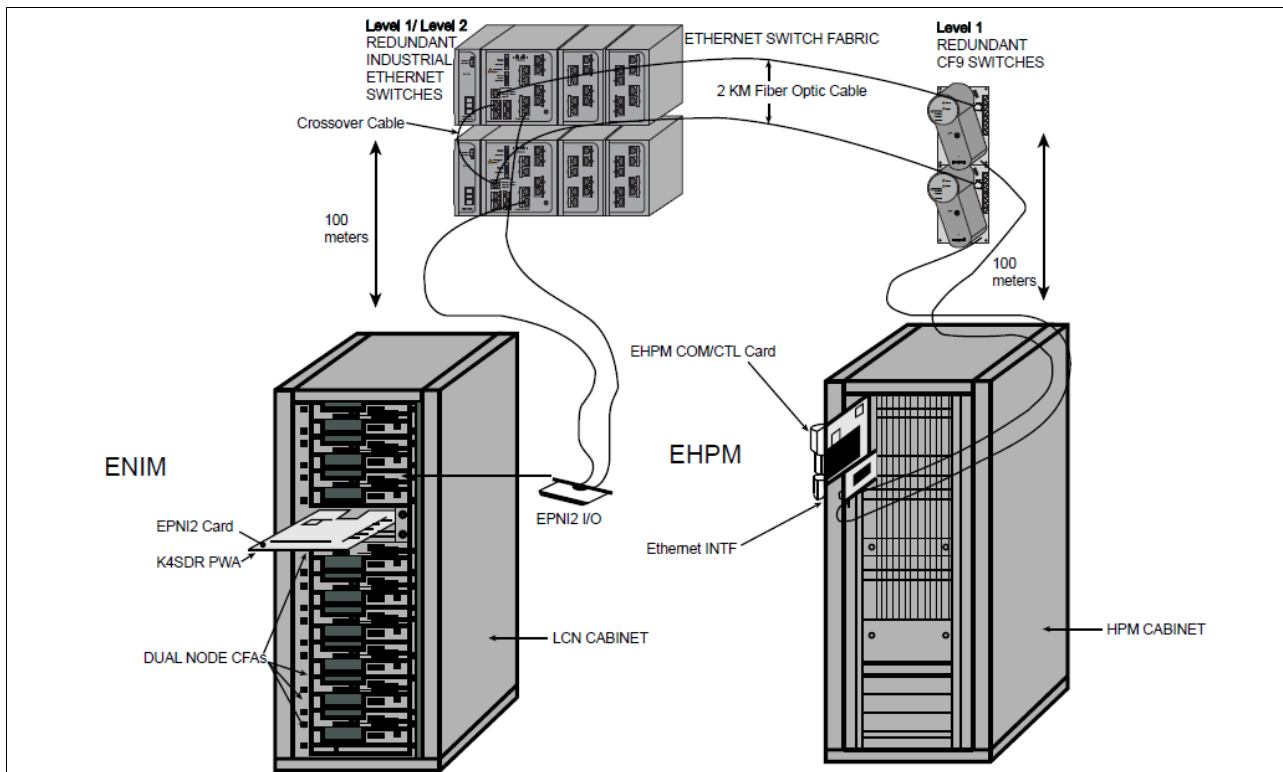


## 8.2 Understanding EUCN network topology

### Typical EUCN Topology

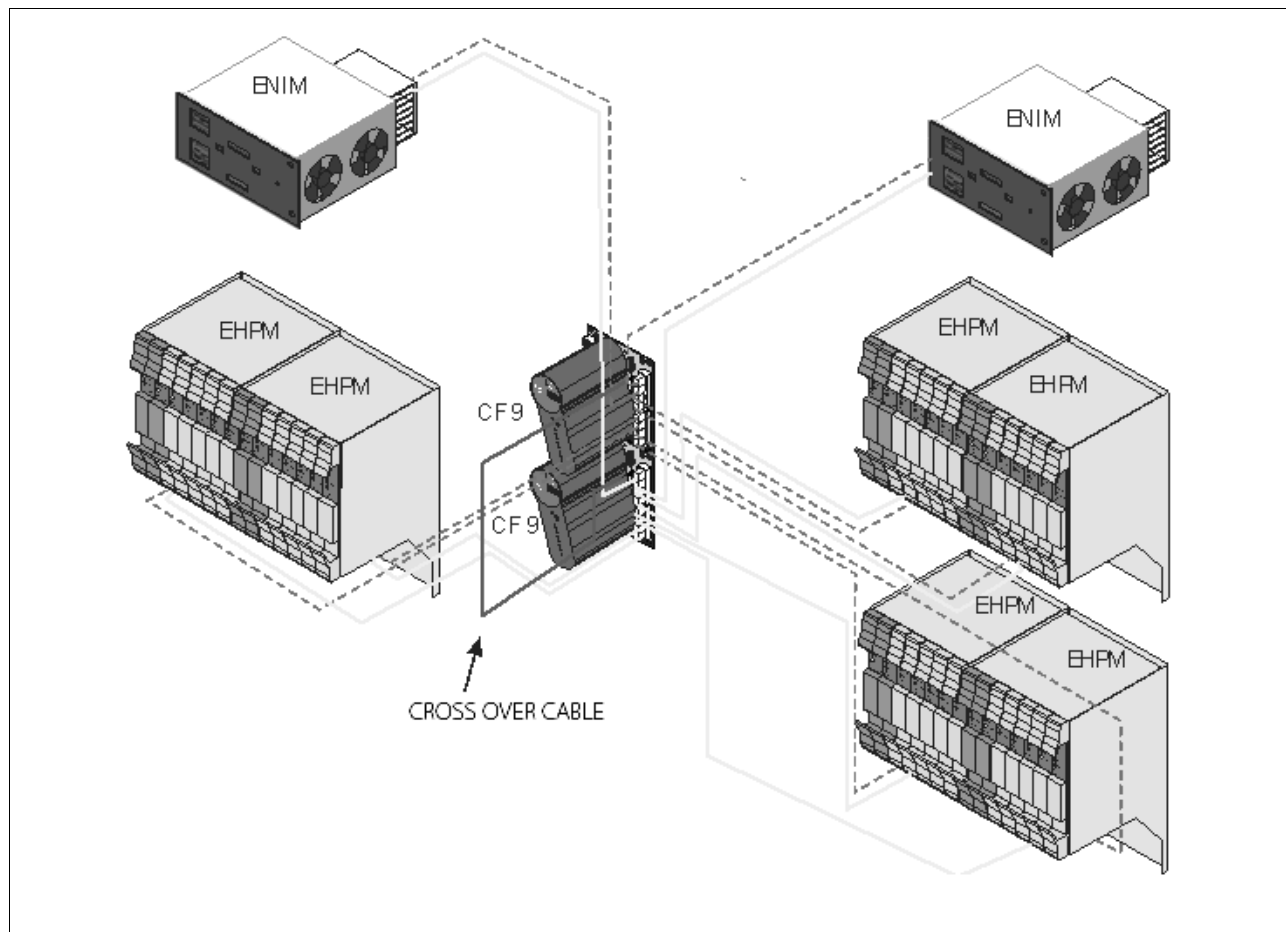
The topology of the FTE based EUCN communication network depends on the distance between the ENIM and the EHPM modules. Generally, a EUCN network consists of ENIM and EHPM modules interconnected with FTE cables, fiber optic cables, and a pair of Ethernet and/or Honeywell Control Firewall 9, which are redundant. Figure 8-2 illustrates a typical network topology. The maximum distance between the EHPM and the ENIM modules is 2 km (1.24 miles); connected using a fiber optic cable. The topology varies with the number of modules that is included in a EUCN network and the distance between them.

Figure 8-2 Typical EUCN Topology



**Basic EUCN Topology** A basic EUCN network consists of two nodes – the ENIM and the EHPM – connected using a Honeywell Control Firewall 9 . See Figure 8-4.

Figure 8-3 Basic EUCN Network Topology



## 8.3 Hardware Components

### Introduction

The various hardware components that compose the EUCN are as follows:

- EPNI2 and EPNI2 I/O cards
- EHPM Comm/Control card and FTE Interface Module Assembly
- Cisco IE 3000 and Honeywell Control Firewall 9
- FTE (A and B) and Fiber optic cables
- Crossover cable

### EPNI2 and EPNI2 I/O cards

The EPNI2 and EPNI2 I/O cards in ENIM replace the EPNI card and the NIM module in NIM. The EPNI2 provides a gateway function between the LCN and the EUCN.

Figure 8-4 EPNI2 card

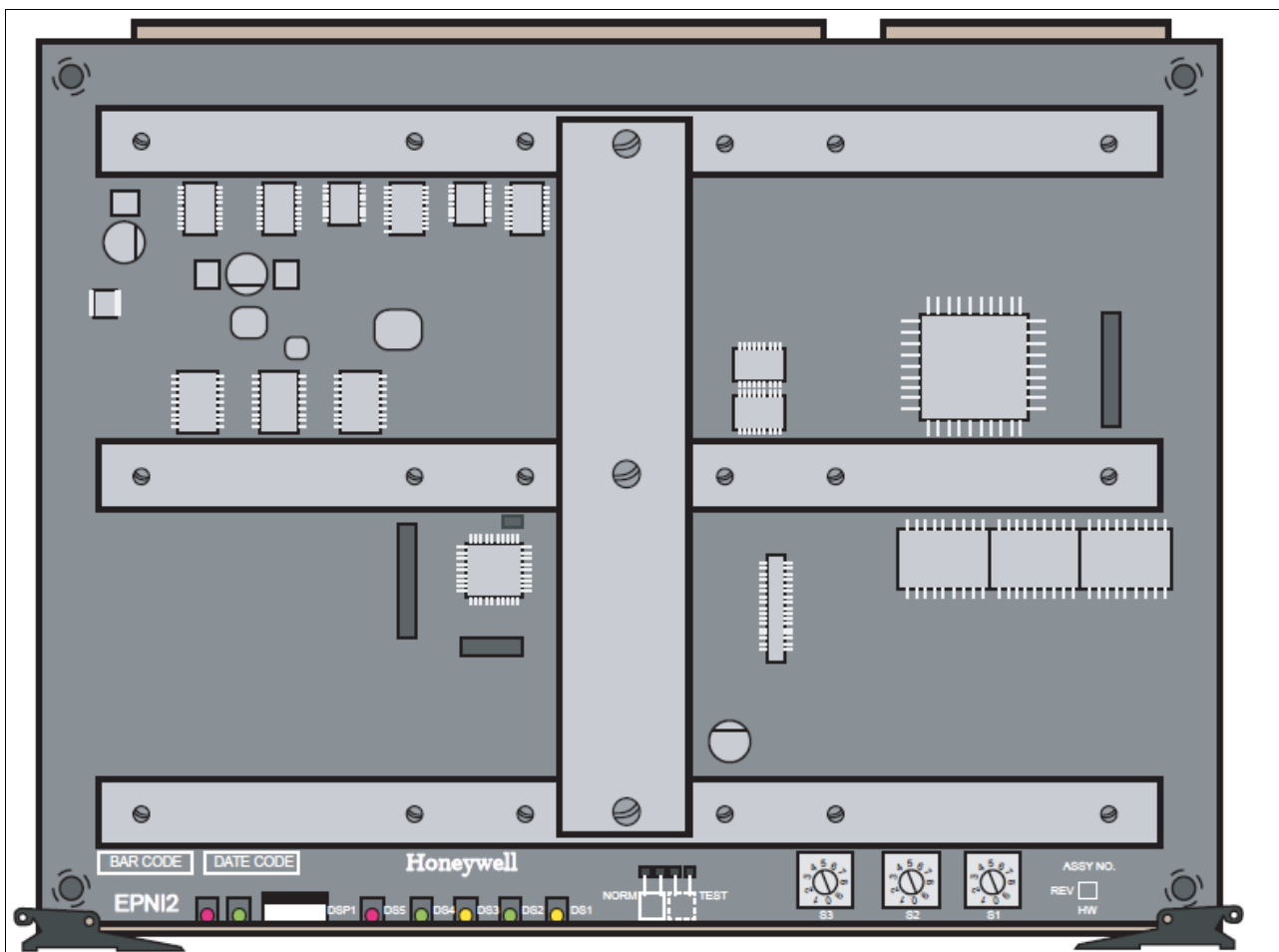
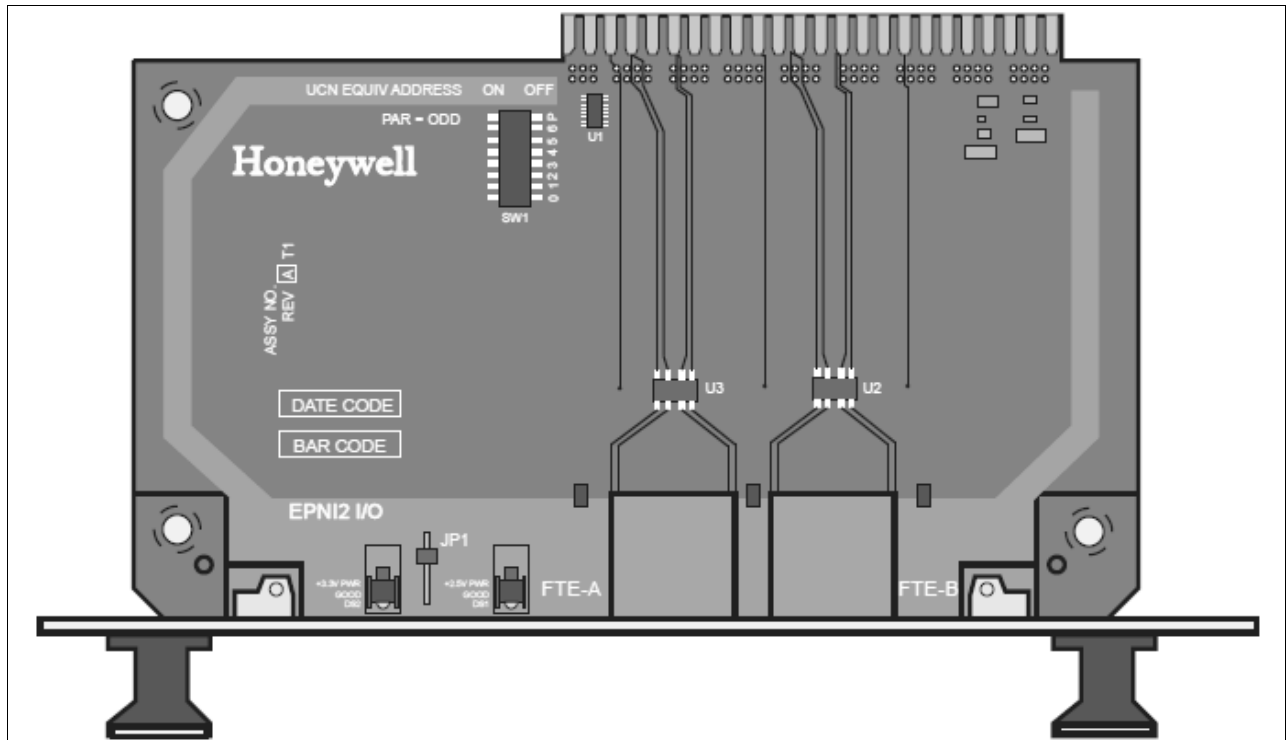


Figure 8-5 EPNI2 I/O card



**EHPM Comm/Control card and FTE Interface Module Assembly**

The EHPM Comm/Control card and the FTE Interface Module Assembly replace the HPM Comm/Control card and the UCN Interface module respectively.

Figure 8-6a FTE Interface Module

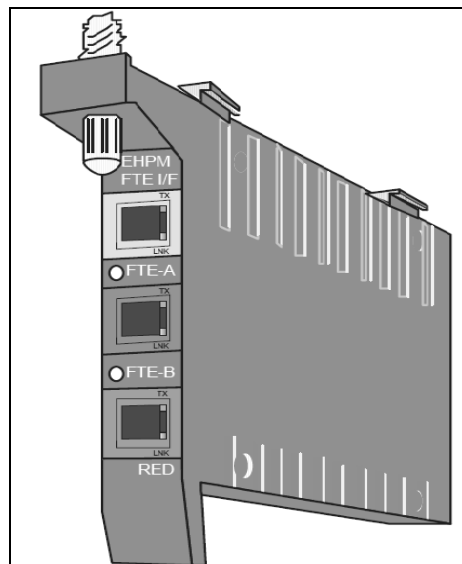
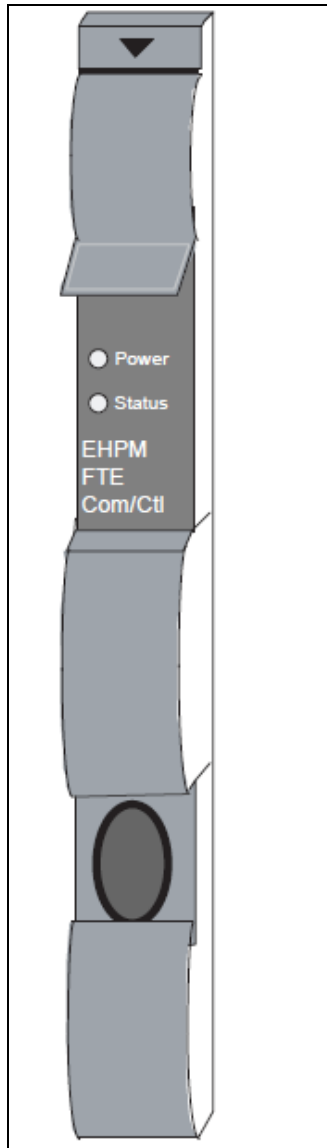


Figure 8-6b EHPM Comm/Control card



---

#### Ethernet Switches

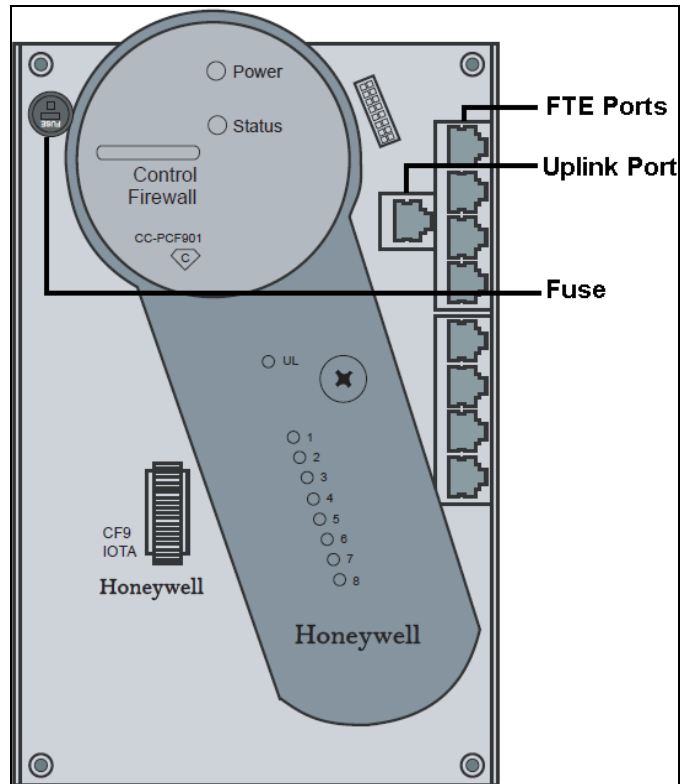
In a EUCN configuration, the following switches are used:

- Honeywell Control Firewall 9
- IE3000 Cisco switches

Honeywell Control Firewall 9 are level 1 switch, which consists of eight FTE port switches and one uplink port. In a basic EUCN topology, it is installed in an EHPM cabinet.

Primary and secondary nodes of a redundant pair of FTE nodes (EHPM, ENIM) must be connected to the same CF9 or switch pair.

Figure 8-7 Honeywell Control Firewall 9



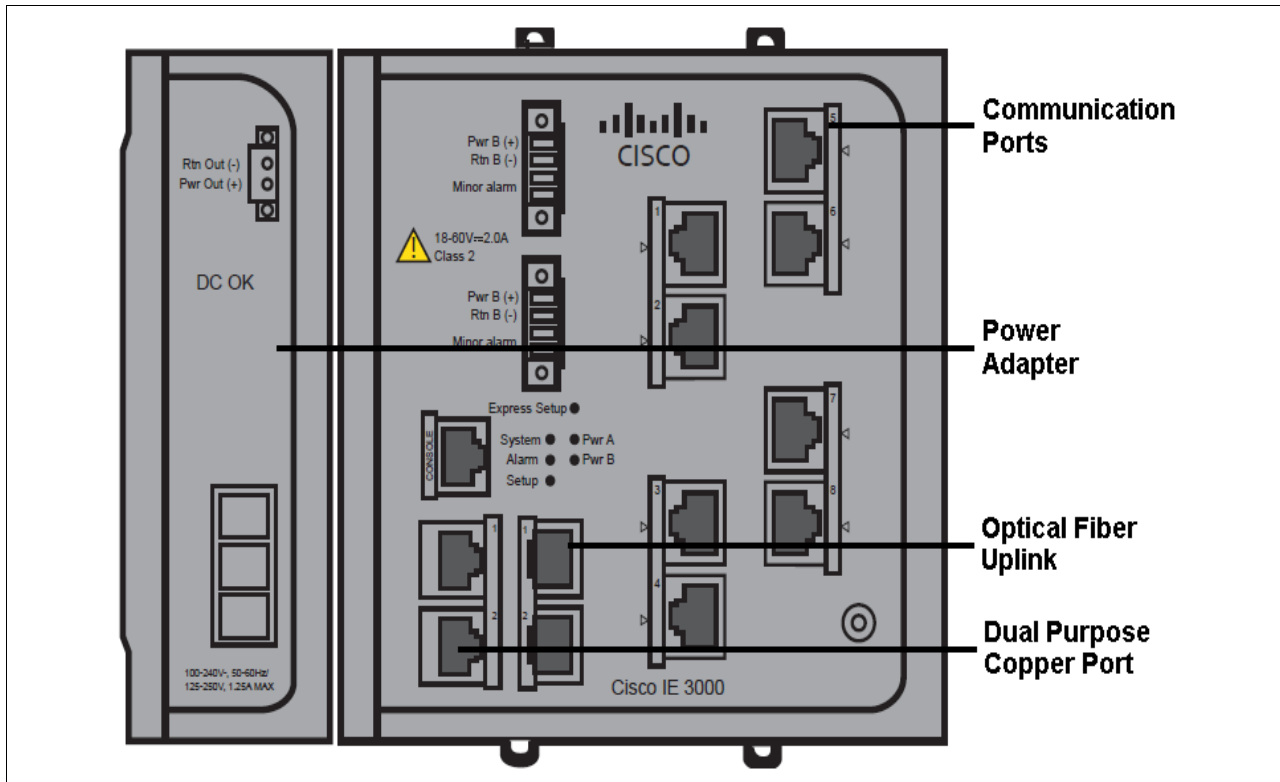
The IE 3000 Cisco switches are a level 2 switch used in a large EUCN network to interconnect the ENIMs and EHPMs. These switches are installed in a LCN cabinet. The IE3000 configuration consists of,

- Maximum number of 8 Level 1 copper ports per switch
- Maximum number of 8 Level 2 copper ports per switch
- Maximum number of 8 Level 2 fiber optic ports per switch (with the inclusion of an 8 port fiber expansion module)

The two Dual Purpose Ports can be configured as,

- extra Level 1 copper ports
- extra Level 2 copper ports
- extra Level 2 fiber optic ports (with additional fiber optic modules) or as Uplink ports or crossover cable ports, as needed.

Figure 8-8 Cisco IE 3000 Ethernet Switch with Power Adapter



#### FTE and fiber optic cables

The FTE cables A and B are distinguished using the color of the dust boots on the cables. The yellow represents the primary FTE (Cable A) and the green represents the secondary FTE (Cable B). Generally, a EUCN network consists of ENIM and EHPM modules interconnected with FTE cables and a pair of redundant Ethernet and/or Honeywell Control Firewall 9.

The maximum “copper cable distance” between the redundant pair of Honeywell Control Firewall 9 /Ethernet switches and any EHPM or ENIM module must be less than or equal to 100 meters. For a cable, distance of more than 100 meters and less than 2 km, fiber optic cables must be used.

#### Crossover cable

In an FTE network, if you are NOT using a router, then only the highest level of redundant Ethernet switches in the LAN are interconnected using the crossover cable.

## Section 9 - Planning an FTE Network

### Section contents

These are the topics covered in this section:

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| 9.3 | Level 1 Best Practices .....     | 83       |
| 9.4 | Level 2 Best Practices .....     | 84       |

### 9.1 Fault Tolerant Ethernet (FTE)

#### Assumptions

Users including network administrators and control engineers, attempting to install and configure an FTE network must understand the FTE networking concepts and requirements, including network design, maintenance, and security. EUCN is a subset of FTE Networks. As a subset, EUCN has a more restrictive set of rules and requirements.

#### Tasks for planning an FTE network for EUCN

Consider the following network requirements before installing your FTE network.

Table 9-1 EUCN/FTE Network Planning Tasks

| ✓ | Task                                                                                                                                                                                                              |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | Be familiar with the FTE topology, including the maximum number of FTE nodes. (Note EUCN limits the number of nodes to 64.)                                                                                       |
|   | Plan your FTE network including the placement of major components, cable segment lengths and limits, and cable routing.                                                                                           |
|   | Understand the security and communication requirements for each level or layer within the FTE network.<br><br>For TPN R684 standalone systems, only two levels are required on EUCN systems, Level 1 and Level 2. |
|   | Plan the use of firewalls, if necessary.                                                                                                                                                                          |
|   | Establish a subnet or domain for your FTE network.                                                                                                                                                                |
|   | Determine all the network settings, including the FTE Nodes' IP addresses.                                                                                                                                        |
|   | Verify the software and media requirements.                                                                                                                                                                       |
|   | Plan the IP address distribution.                                                                                                                                                                                 |

## 9.2 FTE Network Infrastructure

---

### Introduction

An FTE network consists of different node types and network devices. This section describes the considerations and requirements for connecting and configuring the network devices to provide a system with significant security and reliability improvements over a simple Ethernet network.

---

### Plant Network levels

A plant network for EUCN consists of four layers or levels. Level numbers simplifies the description of the node location within the network hierarchy.

The FTE network of a EUCN system includes the following levels.

- Level 2 Supervisory Control, Operator HMI (HMI, and Supervisory Controllers)
  - Level 1 Real Time Control (Controllers and IO)
- 

### FTE communities

An FTE community is a group of nodes with fault tolerant communication coverage that uses FTE test messages. The FTE community uses a common multicast address for the FTE test messages. These nodes are members of the same broadcast domain. Nodes that are single attached or are dual attached but do not run FTE are not members of the FTE community. However, they may communicate on the same network. Honeywell recommends each FTE community to be in a separate broadcast domain.

EUCN is a subset of an FTE community. Presently, the subset of FTE that makes up EUCN is limited to ENIMS, EHPMs, Control Firewalls, and Honeywell recommended Industrial Ethernet switches. Single attached or dual attached (Non-FTE) nodes are not permitted.

---

### Maximum nodes within an EUCN/FTE Community

Each EUCN authentication group is limited to a maximum of 64 nodes. The node limit for EUCN remains the same as in UCN.

---

## 9.3 Level 1 Best Practices

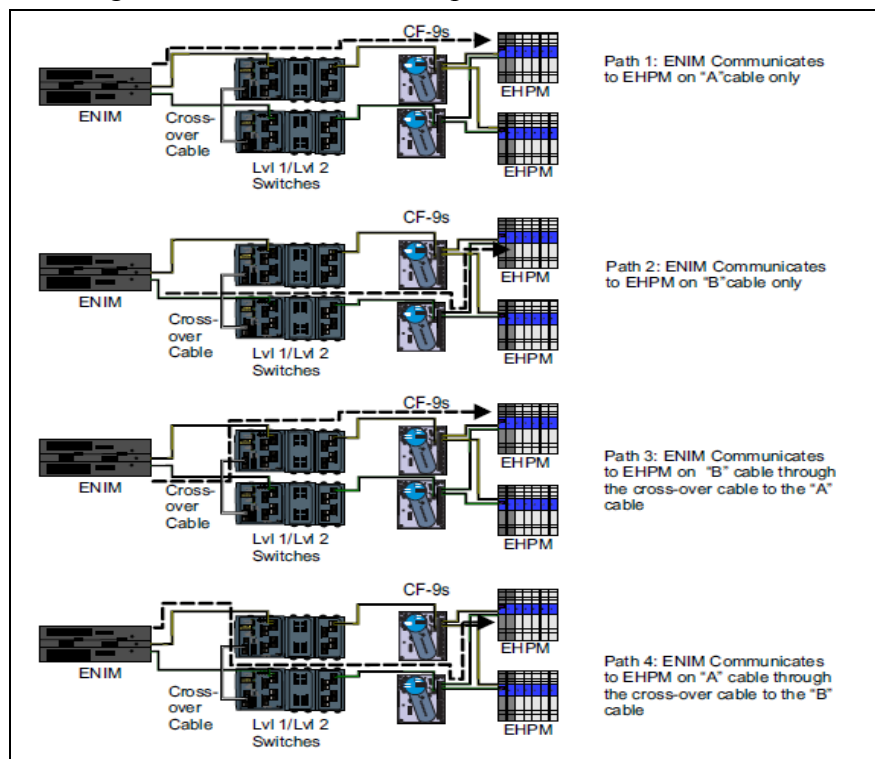
The Level 1 best practice is to place the Level 1 nodes on a separate switch pair or on a Honeywell Control Firewall pair. EHPM nodes **MUST** be connected to Honeywell Control Firewalls. This allows for a critical peer-to-peer traffic that cannot tolerate a communication delay of longer than 250 ms following an FTE cable fault. In addition, it gives the controllers a level of isolation from the other nodes during a catastrophic failure or network disturbance. Controller nodes are connected to level 1 and require adequate bandwidth for critical control traffic. Hence, while identifying the components to be connected for level 1 switch pair, identify the most critical elements of control (EHPMs).

### Connecting Level 1 nodes that intercommunicate

The best practice is to connect the Level 1 nodes that intercommunicate to the same switch pair, to provide the shortest communication path. If this is not possible due to the size or the geographic dispersion, then their communications may go through the Level 2 switches. The Level 2 switches must be configured with the same quality of service approach as those used for Level 1 switch.

Communications redundancy is provided for peer-to-peer traffic by always having two connections (cable “A” and cable “B”) from each node and by using FTE to provide four possible paths (See Figure 9-1).

Figure 9-1 EUCN Multi-path



**Using a switch for level 1 and level 2 (split switch configuration)**

EUCN configurations with a Level 2 switch actually divide the switch into a level 1 and a level 2 section. For instance, a EUCN Cisco IE 3000 Ethernet switch can be used as a split switch. This split switch enables the ENIM to be a member of a Level 1 EUCN/FTE community. The rest of the switch is configured for level 2, to provide connection to the Control Firewall downlink ports.

---

## 9.4 Level 2 Best Practices

---

EUCN networks use Level 2 switches in a qualified configuration. Files for qualified configurations are available. Level 2 ports on EUCN are used for providing connection to the Control Firewalls.

---

**Level 2 switch configuration**

The following are configured in the Cisco switches:

- Protection from broadcast and multicast storms.
  - BPDUguard is configured on non-uplink interfaces to prevent loops and unexpected uplink placement.
- 

**Best practices for connecting a crossover cable**

FTE networks require a single crossover cable at the top of the hierarchy and EUCN networks are no exception to this rule. It is recommended that a 1 Gbps connection be used, because in the case of multiple faults backbone traffic can pass through this connection. Consequently, the highest bandwidth must be available for this traffic.

---

## Appendix A – Worksheets

### A.1 Overview

---

#### Section contents

These are the topics covered in this section:

| Topic |                     | See Page |
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---

### A.2 UCN Worksheets

---

#### Introduction

Copy the sheets on the following pages, fill them out, and then transfer that information to a detailed topology map of your network.

---

## A.2 UCN Worksheets, Continued

### UNIVERSAL CONTROL NETWORK MAP (WORKSHEET)

(Use multiple, consecutive worksheets to show entire network; show terminators at end of trunk)

Customer \_\_\_\_\_ System ID \_\_\_\_\_ Date \_\_\_\_\_

Network Name \_\_\_\_\_

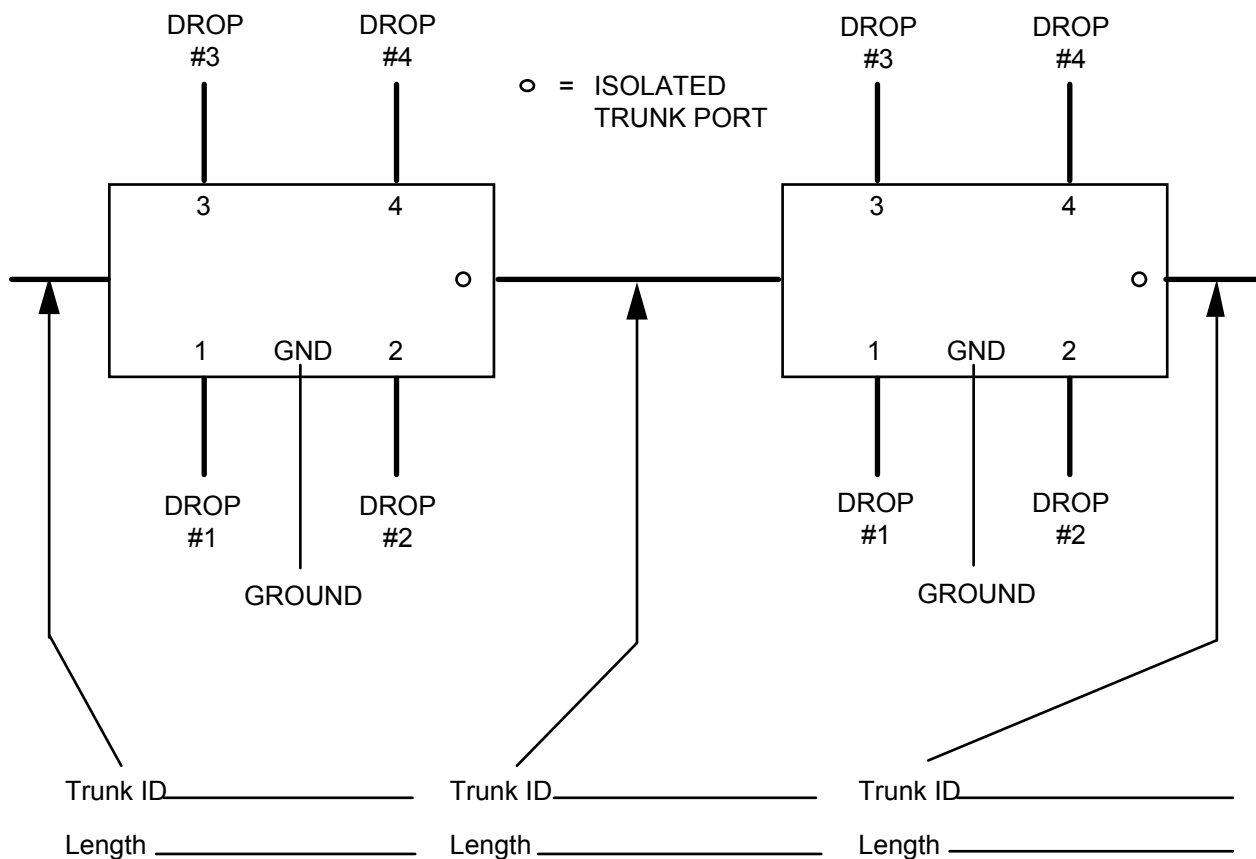
UCN Truck \_\_\_\_\_ (Enter "A" or "B")

Tap type: 2 or 4

Tap ID \_\_\_\_\_

Tap type: 2 or 4

Tap ID \_\_\_\_\_



Drop #1 ID/Length \_\_\_\_\_ Drop #1 ID/Length \_\_\_\_\_

Drop #2 ID/Length \_\_\_\_\_ Drop #2 ID/Length \_\_\_\_\_

Drop #3 ID/Length \_\_\_\_\_ Drop #3 ID/Length \_\_\_\_\_

Drop #4 ID/Length \_\_\_\_\_ Drop #4 ID/Length \_\_\_\_\_

2300

## A.2 UCN Worksheets, Continued

### UNIVERSAL CONTROL NETWORK MAP (WORKSHEET)

(Use multiple, consecutive worksheets to show entire network; show terminators at end of trunk)

Customer \_\_\_\_\_ System ID \_\_\_\_\_ Date \_\_\_\_\_

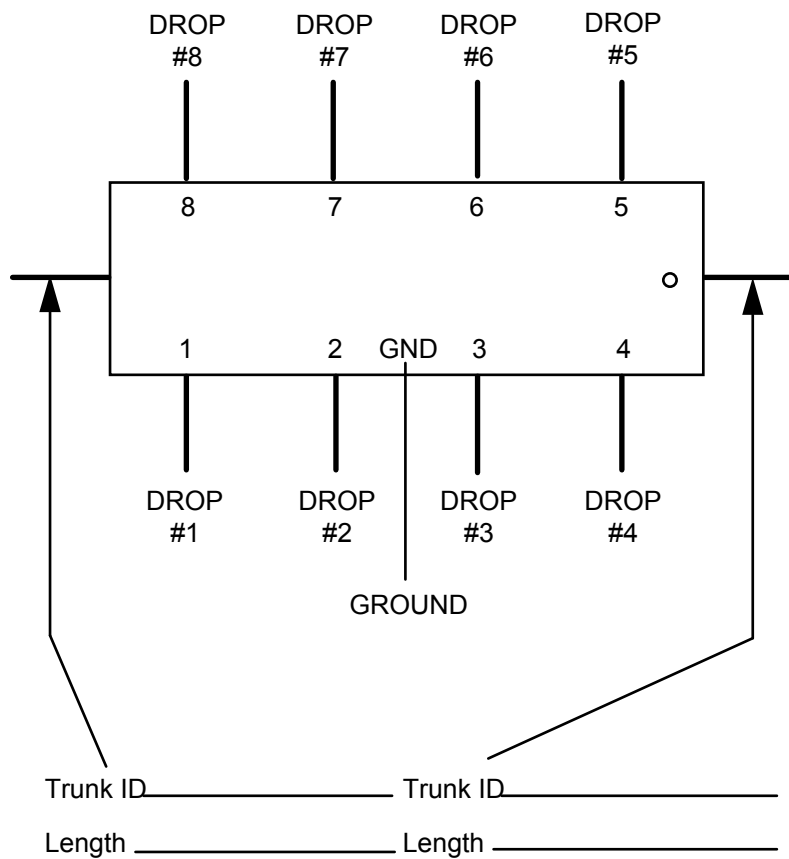
Network Name \_\_\_\_\_

UCN Truck \_\_\_\_\_ (Enter "A" or "B")

Tap type: 8

Tap ID \_\_\_\_\_

○ = ISOLATED  
TRUNK PORT



|                         |                         |
|-------------------------|-------------------------|
| Drop #1 ID/Length _____ | Drop #5 ID/Length _____ |
| Drop #2 ID/Length _____ | Drop #6 ID/Length _____ |
| Drop #3 ID/Length _____ | Drop #7 ID/Length _____ |
| Drop #4 ID/Length _____ | Drop #8 ID/Length _____ |

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Comments: \_\_\_\_\_

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Recommendations: \_\_\_\_\_

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### **FROM:**

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Title: \_\_\_\_\_

Company: \_\_\_\_\_

Address: \_\_\_\_\_

City: \_\_\_\_\_ State: \_\_\_\_\_ ZIP: \_\_\_\_\_

Telephone: \_\_\_\_\_ FAX: \_\_\_\_\_

You may also call 800-822-7673 (available in the 48 contiguous states except Arizona; in Arizona dial 602-313-5558, or write to:

Honeywell Process Solutions  
1860 W. Rose Garden Lane  
Phoenix, AZ 85027 USA





